Turbulence Modelling of Mixing Layers under Anisotropic Strain

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Abstract

The development of turbulent mixing layers can be altered by the application of anisotropic strain rates, potentially arising from radial motion in convergent geometry or movement through nonuniform geometry. Previous closure models and calibrations of compressible turbulence models tend to focus on incompressible flows or isotropic strain cases, which is in contrast to many real flow conditions. The treatment of bulk compression under anisotropic strain is investigated using the K-L turbulence model, a two-equation Reynolds-averaged Navier-Stokes (RANS) model that is commonly used for simulating interfacial instabilities. One-dimensional simulations of shock-induced turbulent mixing layers under applied axial or transverse strain rates are performed using three different closures for the bulk compression of the turbulent length scale. The default closure method using the mean isotropic strain rate is able to reasonably predict the integral width and turbulent kinetic energy of the mixing layer under the applied strain rates. However, the K-L model's performance is improved with the transverse strain closure, while the axial strain closure worsens the model. The effects of this new closure are investigated for the buoyancy-drag model, showing that a three-equation model which evolves the integral width and turbulent length scale separately is most effective for modelling anisotropic strain.

I. INTRODUCTION

Strain rates in fluid flow are a key driver of turbulence, responsible for the shear production of turbulent kinetic energy and the vortex stretching of eddies. Large scale mean strain rates are observed under a variety of scenarios, with shear strain occurring in boundary layers and normal strain occurring at geometric deformations such as nozzles. The evolution of turbulent mixing layers under mean normal strain is not as well studied as shear driven mixing layers, such as those induced by the Kelvin-Helmholtz instability. In the high strain rate limit, where the strain timescale is much smaller than the turbulence timescale, the evolution of turbulence is described by rapid-distortion theory [1–4]. These studies provide useful tools for understanding the pressure-dilatation tensor, however many are limited by the assumption of homogenous turbulence. This is in contrast to realistic applications where turbulence is inhomogeneous and may not satisfy the high strain limit. Likewise,

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the analysis of incompressible fluid flow is simpler than the analysis of compressible flow, with many compressible flow models relying upon incompressible flow closures. For example, several compressible closures have been proposed for the pressure-dilatation term in the transport equation for the turbulent kinetic energy [5–8], however it is common to use the incompressible closure which removes the net contribution of pressure-dilatation [9–14].

Two prominent interfacial instabilities that give rise to turbulent mixing layers are the Rayleigh-Taylor instability (RTI) and the Richtmyer-Meshkov instability (RMI). These instabilities arise from the deposition of vorticity from baroclinic production at the interfaces between fluids due to the misalignment of pressure and density gradients. RTI refers to the case of a persistent pressure gradient, causing one fluid to accelerate into the other, however RTI is only unstable for the light fluid accelerated toward the heavy, and stable for the reverse arrangement. RMI refers to the case of an impulsive acceleration, such as occurs for a shock-wave traversing through the interface. The deposition of vorticity occurs over a far shorter timespan, with RMI being unstable for both heavy-to-light and light-to-heavy shock passages. The linear regime, for which the interface amplitude is much smaller than the wavelength perturbation, is well understood for these instabilities, growing exponentially and linearly for RTI and RMI respectively. In convergent geometry, alterations to the amplitude growth rate are described by the Bell-Plesset effects, which introduce dependencies on the fluid compression rate and convergence rate of the interface radius [15–18]. The aforementioned compression rate and convergence rate can be rewritten in terms of the radial/axial and circumferential/transverse strain rates [19, 20], and so the effects of anisotropic strain rates on the linear regime is within the scope of analytical evaluation. The late-time behaviour of the mixing layer is non-linear, as the interface amplitudes begin to saturate and roll up due to the Kelvin-Helmholtz instability. The instabilities transition to a turbulent mixing layer, with widths growing quadratically ($\sim t^2$) and sub-linearly ($\sim t^\theta$ where $\theta \in (0, 1)$) for RTI and RMI respectively. These mixing layers are anisotropic and inhomogeneous, however for high-Reynolds numbers they are able to achieve self-similar growth/decay states. A thorough review of the RMI and RTI can be found in Refs. [21–23].

In spherical configurations, implosions and explosions are examples of bulk fluid moving to inner and outer radii, respectively. A turbulent mixing layer present in an implosion or explosion will experience mean strain rates due to the radial velocity profile. The mean radial strain rate ($\bar{S}_{rr}$) and circumferential strain rate ($\bar{S}_{\theta\theta}$) for a spherically symmetric flow are given by

$$\bar{S}_{rr} = \frac{\partial \bar{u}_r}{\partial r}, \tag{1}$$

$$\bar{S}_{\theta\theta} = \frac{\bar{u}_r}{r}. \tag{2}$$

These strain rates depend upon the mean radial velocity ($\bar{u}_r$) and the radius (r).

Hydrodynamic instabilities and the advection of fluids play an important role in the evolution of stellar bodies [24]. For example, core-collapse supernovae are some of the largest explosions in the universe, with mixing occurring due to an expansive blast-wave from the core that induces both RMI and RTI [25]. An idealised version of this problem is given by the Taylor-von Neumann-Sedov blast-wave, which provides a self-similar analytical solution for a spherically expanding blast-wave [26–29]. The self-similar profiles are functions of the radius relative to the time-varying radial position of the shock front of the blast-wave, $\zeta = r/R(t)$. The velocity profile scales proportionally to the self-similar function $V(\zeta)$, which is typically solved for numerically. The velocity profile,

$$v(r,t) = \frac{2rV(\zeta)}{5t} \tag{3}$$

describes a self-similar profile as shown in Fig. 1, where the velocity is normalised by the velocity of the fluid immediately behind the shock-wave, v_1. The velocity profile can be used to obtain the radial and circumferential strain rates. The anisotropy of the strain rates is also self-similar as a function of ζ:

$$\bar{S}_{rr}/\bar{S}_{\theta\theta} = 1 + \frac{\zeta}{V(\zeta)}\frac{dV(\zeta)}{d\zeta}. \tag{4}$$

This equation is plotted in Fig. 1 and shows a profile that deviates away from the isotropic value of unity. Immediately behind the shock-front the strain rates are anisotropic, with the radial strain rate up to twice as large as the circumferential strain rate. Whilst this model does not explicitly consider the interaction of the blast-wave with an interface and the resulting wave transmission and reflection, inferences can still be made about the evolution of the mixing layer. The interaction between the blast-wave and an interface between different density fluids will induce RMI due to the blast-wave's impulsive acceleration. Following the blast-wave interaction, the mixing layer's relative radius (ζ) will decrease from $\zeta = 1$, and experience anisotropic strain rates, stretching more in the radial direction than the circumferential direction. As time continues, the strain rates experienced by the mixing layer will tend toward isotropy. Whilst there is also a pressure gradient following the blastwave which can induce RTI, the effects of the buoyancy production under strain rates is not the focus of this work.

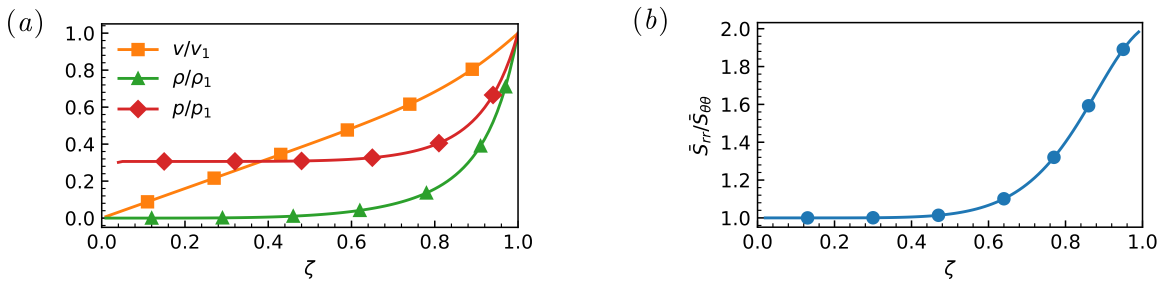


FIG. 1: Self similar profiles as a function of relative radius, $\zeta = r/R(t)$, for the Taylor-von Neumann-Sedov blastwave. (a) Velocity, density, and pressure normalised to values immediately behind the shock front; (b) Strain rate anisotropy.

Inertial confinement fusion is an example of an implosion profile, where small capsules containing fuel for nuclear fusion are compressed [30], driven directly by lasers, or indirectly by X-rays [31]. To achieve the pressures and temperatures necessary for fusion to occur, a series of pulses are used to launch shock waves into the capsule, causing the capsule to compress. Perturbations in the interfaces between the capsule's layers can arise from limitations in machining, or from drive asymmetries on the capsule surface. These perturbations make the system susceptible to hydrodynamic instabilities, either from shock interaction (RMI) or interface accelerations (RTI). Additional shock waves can amplify the existing instabilities, further accelerating the growth of the interface perturbations. These instabilities are responsible for mixing the cold, outer fuel with the hot fuel at the centre hot-spot, degrading performance and preventing ignition [32, 33]. A simplified, idealised implosion profile used for modelling the hydrodynamics of inertial confinement fusion was presented in Ref. [34] and used as a template for many further investigations for both single-mode [35, 36] and multi-mode interface perturbations [37–41]. The mixing layer position and the mean radial velocity profile at several time instants are shown in Fig. 2 for the implosion profile, using the data from Ref. [41]. Evaluating the displayed mean radial velocity profiles, the radial velocity gradient across the mixing layer shows a positive trend indicating expansive radial strain rates. The radial velocity is negative, as expected for an implosion, which correlates to a compressive circumferential strain rate. These profiles are taken during the intervals

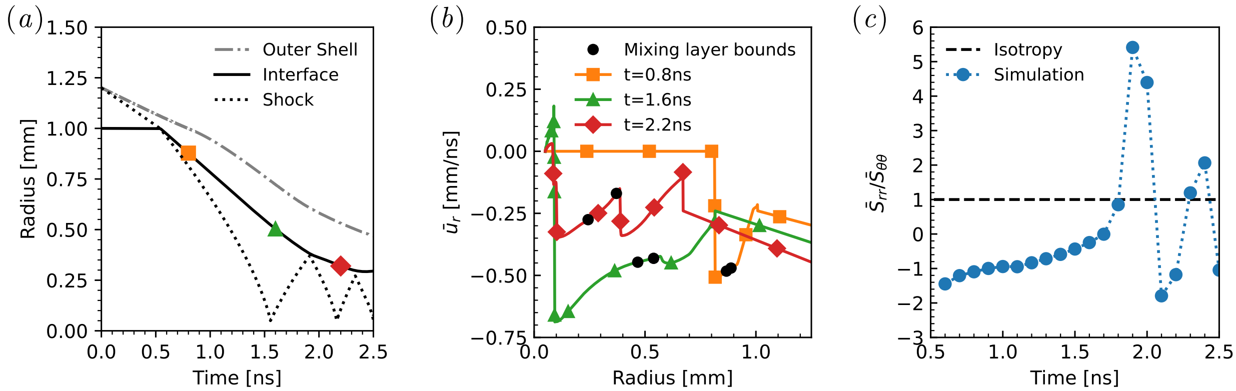


FIG. 2: Characteristics of idealised implosion simulation [41]. (a) Implosion profile, (b) mean radial velocity profile at several time-steps, (c) strain rate anisotropy across the mixing layer.

between shock interactions, and suggest that when the mixing layer is evolving between shocks it experiences anisotropic strain rates. The anisotropy of the strain rates across the mixing layer is also plotted in Fig. 2, and shows that the ratio of the strain rates tends towards -1 rather than unity. There are times at which the anisotropy ratio goes above unity, aligning with the times the shock wave is compressing the turbulent mixing layer, such that the radial strain rate is more compressive than the circumferential strain rate. However, for the majority of the simulation the circumferential strain rate is negative (compression), whilst the radial strain rate is positive (expansive).

Reynolds-Averaged Navier-Stokes (RANS) models are an important tool to used to predict the behaviour of hydrodynamic instabilities at a fraction of the cost of large-eddy simulation (LES) and direct numerical simulation (DNS). For practical problems such as astrophysical flows and ICF, it can be computationally intractable to completely resolve the complete range of length and time scales involved, driving the necessity for simplified models. For compressible turbulent mixing layers, Besnard *et al.* [42] presented a comprehensive derivation of the Favre-averaged equations for variable-density turbulence. In Ref. [42], both a Reynolds stress model and a simpler three equation model were provided, forming the basis of the BHR model. Both models included an evolution equation for the turbulent kinetic energy dissipation rate ϵ. Using the dissipation rate is a popular choice in other fields, given the prominence of the K-ϵ turbulence model [9, 43, 44], and several other turbulence models designed for compressible turbulent mixing layers have also used the turbulent dissipation rate [11, 13, 45]. Whilst there is an equivalence between the transport equations, an alternative approach is to transport the turbulent length scale L, as recommended in Ref. [42] and implemented by Dimonte and Tipton [12] for the K-L model. For interfacial instabilities such as RTI and RMI, the initial turbulent length scale has a greater physical meaning than an initial dissipation rate, and also allows for easier self-similarity analysis at late-time where the turbulent length scale is assumed to be proportional to the mixing layer width. Improvements have been made to the K-L model to improve performance, through self-similarity analysis [46–49], improved treatment of buoyancy production for the turbulent kinetic energy [50], and realisability of the modelled Reynolds stress [51, 52]. Further developments to the BHR model and K-L model have occurred through the introduction of additional transport equations resulting in a multitude of different models [53–62]. One of the lesser analysed components of the closure for the RANS models is the effect of bulk compression. The recommended modification by Reynolds [63, 64] for the dissipation rate based upon isotropic compression in the rapid distortion regime for homogenous turbulence is equivalent to the term by Dimonte and Tipton [12] for the turbulent length scale derived for conservation of mass of an eddy under isotropic compression [65]. Investigating isotropic compression of homogenous turbulence for sudden viscous dissipation, Campos and Morgan [66] used the same expression for bulk compression but found it necessary to introduce a variable viscosity term in the length scale transport equation to align with DNS results. As noted previously, in practical applications turbulent mixing layers are neither homogenous or experiencing isotropic strain.

The goal of this paper is to quantify the ability of the K-L turbulence model to capture the characteristics of turbulent mixing layers under anisotropic strain rates and propose an improved model formulation. The work focuses on the bulk compression term for the scaling of the turbulent length scale under strain. Section II introduces the equations of the K-L model, the method used to apply the uniform strain rates to the simulation domain, and the simulation cases conducted. In Sec. III, the performance of the K-L model with different scaling approaches is presented, showing results for the integral width and the domain integrated turbulent kinetic energy. Further insight into the K-L model is obtained by performing a self-similar analysis of the K-L model to reduce it down to a buoyancy-drag model. The conclusions are summarised in Sec. IV.

II. COMPUTATIONAL APPROACH

A. Strain rates

In order to analyse the influence of the strain rates on the development of the turbulent mixing layer, it is necessary to quantify and control the strain rate. The application of uniform normal strain rates are achieved by applying mean velocity gradients to the simulation domain and moving the domain bounds with the specified strain rates, using boundary conditions which preserve the linear velocity profile. For planar geometry, the strain rates are considered separately in the axial direction ($\bar{S}_A$), representing strain rates in the direction of mixing for the inhomogeneous turbulent mixing layer, and in the transverse direction ($\bar{S}_T$), corresponding to strain rates in the directions of the homogeneous plane.

$$\bar{S}_A = \frac{\partial \bar{u}_1}{\partial x_1} \tag{5}$$

$$\bar{S}_T = \frac{\partial \bar{u}_2}{\partial x_2} = \frac{\partial \bar{u}_3}{\partial x_3} \tag{6}$$

A combination of axial and transverse strain is possible, such as for isotropy ($\bar{S}_A = \bar{S}_T$) or for bulk incompressibility with zero divergence ($\bar{S}_A + 2\bar{S}_T = 0$).

Two different strain-profiles are utilised. The first is the constant velocity profile, which is the default strain-profile used as it satisfies the momentum equation with uniform pressure. The constant velocity profile corresponds to the case where the domain extent expands/compresses with a constant velocity. The strain rate of this profile varies with time, which can be calculated from the initial strain rate,

$$\bar{S}(t > t_0) = \frac{\bar{S}_0}{1 + \bar{S}_0(t - t_0)} \tag{7}$$

where $\bar{S}_0$ is the strain rate at an initial time of t_0. The second strain profile applies a constant strain rate in time. This is achieved by including a source term to help accelerate the flow to the desired profile, causing the domain extent to expand/compress exponentially. The source term allows growth without a pressure gradient, thus avoiding the Rayleigh-Taylor instability and isolating the influence of the strain rate on the RMI-induced turbulence.

The strain rates are non-dimensionalised by the inverse of the factor used for the nondimensionalisation of time:

$$\tau = \frac{t\dot{W}_0}{\bar{\lambda}} \tag{8}$$

$$\hat{S} = \frac{\hat{S}\bar{\lambda}}{\dot{W}_0} \tag{9}$$

The non-dimensional time, τ, is calculated from the initial eddy turnover time for the RMI-induced mixing layer which depends on the initial linear growth rate of the integral width, $\dot{W}_0$, and the initial mean wavelength of the mixing layer, $\bar{\lambda}$. This linear transformation preserves the relation $\bar{S}t = \hat{S}\tau$, which is useful for calculating the expansion of the domain as denoted by the expansion factor,

$$\Lambda(t) = \exp\left[\int_0^t \bar{S}(t')dt'\right] = \exp\left[\int_0^\tau \hat{S}(\tau')d\tau'\right] \tag{10}$$

which gives the change in length ratio in the direction of the specified strain rate.

B. Governing Equations

The simulations conducted use the K-L model, a two-equation RANS model that achieves closure by introducing transport equations for the turbulent length scale L and turbulent kinetic energy $\tilde{K} = \widetilde{u_i''u_i''}/2$. The model is derived from the compressible, multi-component Navier-Stokes equations using Favre averages. The Reynolds average decomposition of a scalar is given by,

$$\phi = \bar{\phi} + \phi' \tag{11}$$

where $\bar{\phi}$ is the Reynolds average and ϕ' is the zero-mean fluctuation, $\overline{\phi'} = 0$. The Favre average is a density weighted average defined by,

$$\widetilde{\phi} = \overline{\rho\phi}/\bar{\phi} \tag{12}$$

that allows a decomposition into the form,

$$\phi = \widetilde{\phi} + \phi'' \tag{13}$$

with a fluctuation term that does not have a zero mean, $\overline{\phi''} \neq 0$, instead it has a zero-mean when density weighted, $\overline{\rho\phi''} = 0$. The governing equations of the K-L model are given by:

$$\frac{\partial \bar{\rho}}{\partial t} + \frac{\partial}{\partial x_j}\left(\bar{\rho}\tilde{u}_j\right) = 0 \tag{14a}$$

$$\frac{\partial \bar{\rho}\tilde{u}_i}{\partial t} + \frac{\partial}{\partial x_j}\left(\bar{\rho}\tilde{u}_j\tilde{u}_i + \bar{p}\delta_{ij}\right) = \frac{\partial \bar{\tau}_{ij}}{\partial x_j} + \bar{\rho}g_i \tag{14b}$$

$$\frac{\partial \bar{\rho}\tilde{E}}{\partial t} + \frac{\partial}{\partial x_j}\left(\left(\bar{\rho}\tilde{u}_j + \bar{p}\right)\tilde{u}_j\right) = \frac{\partial}{\partial x_j}\left(\frac{\mu_T}{N_H}\frac{\partial \tilde{h}}{\partial x_j} + \frac{\mu_T}{N_K}\frac{\partial \tilde{K}}{\partial x_j}\right) + \frac{\partial \bar{\tau}_{ij}\tilde{u}_i}{\partial x_j} + \bar{\rho}g_i\tilde{u}_i \tag{14c}$$

$$\frac{\partial \bar{\rho}\tilde{Y}_a}{\partial t} + \frac{\partial}{\partial x_j}\left(\bar{\rho}\tilde{Y}_a\tilde{u}_j\right) = \frac{\partial}{\partial x_j}\left(\frac{\mu_T}{N_Y}\frac{\partial \tilde{Y}_a}{\partial x_j}\right) \tag{14d}$$

$$\frac{\partial \bar{\rho}\tilde{K}}{\partial t} + \frac{\partial}{\partial x_j}\left(\bar{\rho}\tilde{u}_j\tilde{K}\right) = \frac{\partial}{\partial x_j}\left(\frac{\mu_T}{N_K}\frac{\partial \tilde{K}}{\partial x_j}\right) + \bar{\tau}_{ij}\frac{\partial \tilde{u}_i}{\partial x_j} + S_K - C_D\bar{\rho}\frac{\tilde{V}^3}{L} \tag{14e}$$

$$\frac{\partial \bar{\rho}L}{\partial t} + \frac{\partial}{\partial x_j}\left(\bar{\rho}\tilde{u}_jL\right) = \frac{\partial}{\partial x_j}\left(\frac{\mu_T}{N_L}\frac{\partial L}{\partial x_j}\right) + C_L\bar{\rho}V + C_C\bar{\rho}L\frac{\partial \tilde{u}_j}{\partial x_j} \tag{14f}$$

where $\bar{\rho}$ is the mean density, $\tilde{u}_i$ is the Favre-averaged velocity vector, $\bar{p}$ is the mean pressure, $\tilde{E}$ is the Favre-averaged total energy, $\tilde{Y}_a$ is the mass fraction of species a, $\tilde{K}$ is the Favre-averaged turbulent kinetic energy, L is the turbulent length scale, and g_i is the body acceleration vector. The turbulent velocity is defined from the turbulent kinetic energy, $\tilde{V} = \sqrt{2\tilde{K}}$. The equations have been closed by using a turbulent viscosity with the gradient diffusion approximation,

$$\mu_t = C_\mu \bar{\rho} L \sqrt{2\tilde{K}}. \tag{15}$$

The Reynolds stress tensor, $\bar{\tau}_{ij} = -\bar{\rho}\widetilde{u_i''u_j''}$, is defined with the Boussinesq eddy viscosity assumption, using the mean velocity gradients and turbulent kinetic energy. The coefficient C_P is taken to be 2/3 to ensure the trace of the Reynolds stress tensor is equal to the turbulent kinetic energy.

$$\bar{\tau}_{ij} = \mu_t\left(\frac{\partial \tilde{u}_i}{\partial x_j} + \frac{\partial \tilde{u}_j}{\partial x_i} - \frac{2}{3}\frac{\partial \tilde{u}_k}{\partial x_k}\delta_{ij}\right) - C_P\bar{\rho}\tilde{K}\delta_{ij} \tag{16}$$

The model differs from the original K-L model by Dimonte and Tipton [12], instead using the enthalpy diffusion and the equation for total energy conservation given in Ref. [50]. The source term for the turbulent kinetic energy, S_K, is required to differentiate between the different forms of acceleration in RMI and RTI. The method implemented uses the approach by Ref. [48], which is based on the work in Ref. [50]. The source term takes the form,

$$S_K = \begin{cases} C_B \bar{\rho}\tilde{V}\ \max\left(A_{L_i} g_i, 0\right), & \Theta_{g_f} \geq \Theta_{g_m} \\ C_B \bar{\rho}\tilde{V} |A_{L_i} g_{L_i}|, & \Theta_{g_f} < \Theta_{g_m} \end{cases} \tag{17}$$

where $g_{L_i} = -(1/\bar{\rho})\partial\bar{p}/\partial x_i$ is the local acceleration calculated from the pressure gradient, and A_{L_i} is the local Atwood number to be calculated from the local flow field. The flow is considered RT-like if the turbulent acceleration, $\Theta_{g_f} = \sqrt{\tilde{K}}/\Delta t^*$, is greater than the mean-field acceleration, $\Theta_{g_m} = (1/\bar{\rho})|\partial p/\partial x|$. The minimum timescale, Δt^*, is calculated by,

$$\Delta t^* = \min\left(\frac{\min(\Delta x_i) CFL}{|\tilde{u} + \bar{c}|}, \frac{L}{\bar{c}}\right), \tag{18}$$

using the CFL number, the minimum side length of the cell, Δx_i, and the local speed of sound, $\bar{c}$. The local Atwood number is calculated from densities reconstructed at the faces using van Leer's monotonicity principle [50]. The second-order slope for the density within each cell is calculated according to

$$\Delta_i = \left\{ \begin{array}{ll} \text{sign}(\Delta_{i+\frac{1}{2}}) \min(|\Delta_{i-\frac{1}{2}}|, |\Delta_{i+\frac{1}{2}}|), & \text{if sign}(\Delta_{i-\frac{1}{2}}) = \text{sign}(\Delta_{i+\frac{1}{2}}) \\ 0, & \text{otherwise} \end{array} \right\} \tag{19}$$

where $\Delta_{i-\frac{1}{2}} = \bar{\rho}_i - \bar{\rho}_{i-1}$ is the unrestricted difference between the adjacent cells. The reconstructed states at each side of the cells are:

$$\bar{\rho}_i^L = \bar{\rho}_i - \frac{1}{2}\Delta_i, \qquad \bar{\rho}_i^R = \bar{\rho}_i + \frac{1}{2}\Delta_i \tag{20}$$

Using the reconstructed densities at each face, the values from either side of the face are averaged to give the face density:

$$\bar{\rho}_{i+\frac{1}{2}} = \frac{1}{2}\left(\bar{\rho}_i^R + \bar{\rho}_{i+1}^L\right) \tag{21}$$

The local Atwood number is calculated based upon an initial estimate and a steady-state estimate,

$$A_{L_i} = (1 - w_L) A_{0_i} + w_L A_{SS_i}. \tag{22}$$

The local initial Atwood number is calculated from the face-averaged density values, which in one-dimension may be written as,

$$A_{0_i} = \left(\frac{\bar{\rho}_{i+\frac{1}{2}} - \bar{\rho}_{i-\frac{1}{2}}}{\bar{\rho}_{i+\frac{1}{2}} + \bar{\rho}_{i-\frac{i}{2}}}\right)_i \tag{23}$$

The self-similar local Atwood number uses a density gradient calculated from the mean density values at the faces. The one dimensional calculation takes the form,

$$\left(\frac{\partial \bar{\rho}}{\partial x}\right)_i = \frac{\bar{\rho}_{i+\frac{1}{2}} - \bar{\rho}_{i-\frac{1}{2}}}{\Delta x}, \tag{24}$$

$$A_{SS_i} = C_A \frac{L}{\bar{\rho} + L|\partial\bar{\rho}/\partial x|_i} \left(\frac{\partial \bar{\rho}}{\partial x}\right)_i. \tag{25}$$

The transition between the two Atwood number estimates is controlled by the weighting factor $w_L = \min(L/\Delta x, 1)$. The choice of coefficients will be discussed in Sec. II D.

The total energy consists of the internal energy $\tilde{e}$, the Favre-averaged velocity, and the turbulent kinetic energy,

$$\tilde{E} = \tilde{e} + \frac{1}{2}\tilde{u}_i\tilde{u}_i + \tilde{K}. \tag{26}$$

For a mixture of ideal gases with an isothermal closure, the internal energy is taken to relate to the pressure according to,

$$\bar{p} = (\bar{\gamma} - 1)\bar{\rho}\tilde{e} \tag{27}$$

which uses the mixtures specific heat ratio $\bar{\gamma}$. With the specific heat capacity at constant pressure and constant volume for species a given by c_{pa} and c_{va} respectively, the mixture specific heat ratio is calculated by,

$$\bar{\gamma} = \frac{\sum_a \tilde{Y}_a c_{va}}{\sum_b \tilde{Y}_b c_{pb}}. \tag{28}$$

The volume fraction can be calculated using an isobaric closure, resulting in the calculation,

$$\bar{f}_a = \frac{\tilde{Y}_a/m_a}{\sum_b \tilde{Y}_b/m_b} \tag{29}$$

where m_a is the molecular weight of species a.

C. Numerical Implementation

The K-L model is implemented into FLAMENCO, a multi-block structured finite-volume code. The code uses an approach based on the Godunov method [67] to evaluate the inviscid fluxes of the governing equations, corresponding to the spatial fluxes on the left-hand-side of Eq. 14. For the inviscid fluxes, the values are reconstructed at the faces using a nominally fifth-order scheme [68], and modified with a low-Mach correction to ensure proper dissipation rate scaling as the Mach number decreases [69, 70]. The flux is calculated using the HLLC Riemann solver [71]. Viscous fluxes are evaluated using second-order centred differences, and the time-stepping is second-order using a total variation diminishing Runge-Kutta method [72]. To apply the strain rates to the simulation domain, the mesh is moved with the applied strain rate, stretching or compressing according to the theoretical profile. To preserve the uniform strain rate in the flow, moving symmetry planes are applied in the direction of the strain rates, allowing the ghost cells to preserve the specified velocity gradient. The flux calculation is altered using the arbitrary Lagragian-Eulerian approach, modifying the velocity component normal to the face to be the velocity relative to the face velocity for the HLLC Riemann Solver [73].

D. Problem Description

To investigate the capability of the K-L model to capture the effects of the uniform strain rates on the development of the turbulent mixing layer, one-dimensional RANS simulations are conducted with applied mean strain rates. Previous implicit large eddy simulations have been conducted looking at the effect of the normal strain rates on the development of the Richtmyer-Meshkov induced mixing layer [19, 20]. These simulations were based upon the quarter-scale θ-group case [74] which simulated a narrowband multi-mode Richtmyer-Meshkov instability with a $Ma = 1.8439$ shock-wave initialising the instability in a heavy-to-light configuration, conducted at $At = 0.5$ with an ideal gas equation of state for the fluids. The strained simulations were conducted separately for axial strain rates [19], and transverse strain rates [20]. At the time of $\tau = 1$ while the mixing layer is transitioning to turbulence, the strain rates are applied by modifying the velocity profile to achieved the desired velocity gradient, and the domain begins to move with the specified profile to capture the strain and preserve the velocity gradients.

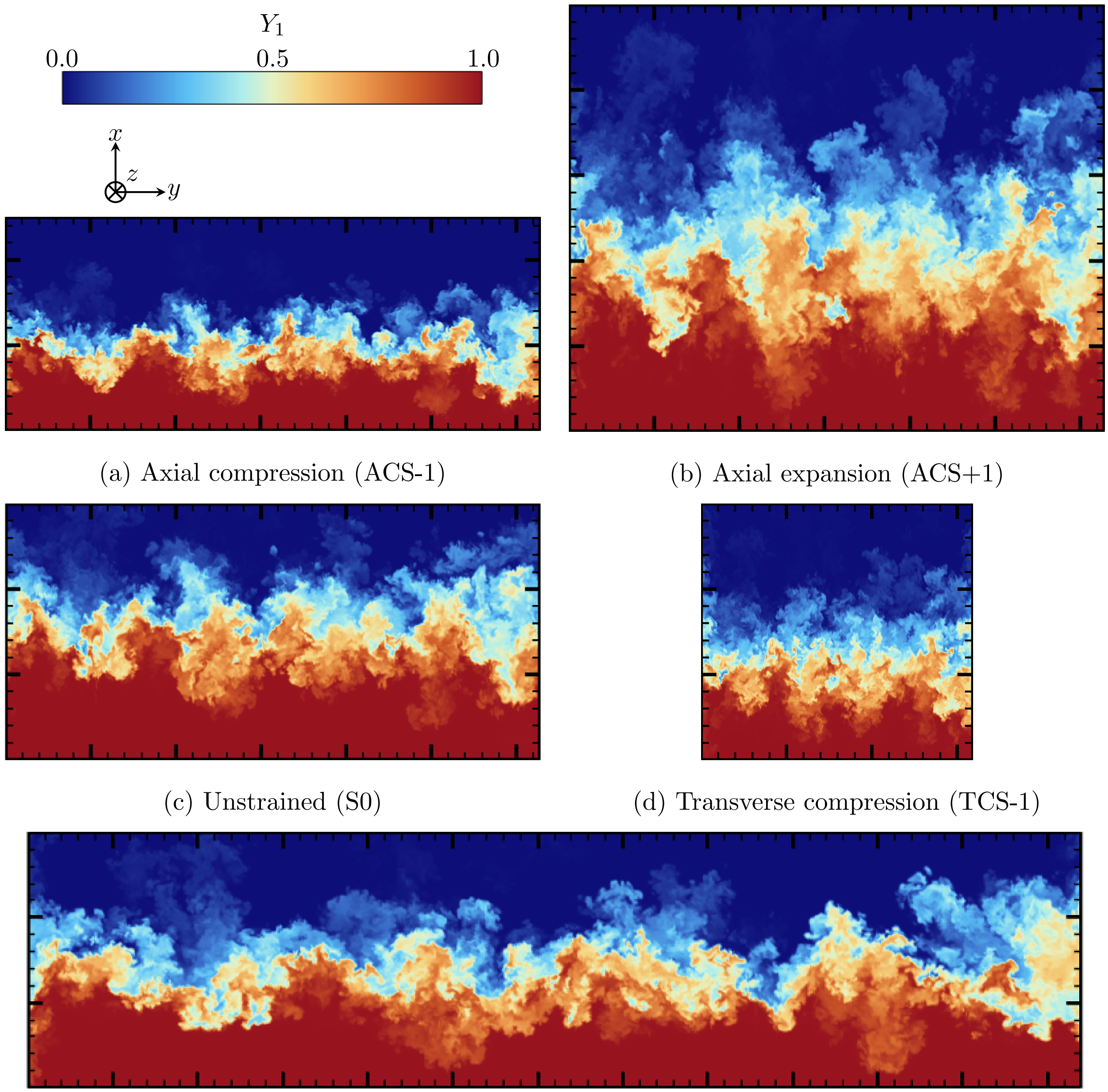
The RANS simulation cases are based on the ILES data in Refs. [19, 20, 74]. The ILES cases consist of an axial strain subset with eight strain cases, and a transverse strain subset with eight cases. These strain subsets use the same strain rate and strain profile combinations for the axial and transverse strain applications. Each case is labelled according to the strain direction, strain profile, and relative strain rate: $D\mathrm{C}X{\pm}n$, where D indicates direction (A for

TABLE I: Properties of the implicit large eddy simulation cases. Simulations cases exist for D=A for the axial strain cases, and D=T for the transverse cases.

		Strain rate		
Label	Strain profile	$\bar{S}$ (s^{-1})	$\hat{S}$	Final τ
S0	Unstrained	0.0	0.0	35
DCV−2	Constant Velocity	−2.5	−0.051	10
DCV−1	Constant Velocity	−0.625	−0.013	35
DCV+1	Constant Velocity	1.25	0.025	35
DCV+2	Constant Velocity	5.0	0.102	10
DCS−2	Constant Strain	−4.00	−0.081	10
DCS−1	Constant Strain	−1.00	−0.020	35
DCS+1	Constant Strain	1.00	0.020	35
DCS+2	Constant Strain	4.00	0.081	10

axial and T for transverse, CX indicates the strain profile (CV for constant velocity and CS for constant strain rate), and $\pm n$ indicates the ranked magnitude of the applied strain rate ($+n$ is expansive, $-n$ is compressive, larger n is larger magnitude strain rate). The specific strain rates for each case and the simulation run time are listed in Table I. Visualisations of the ILES constant strain rate cases at the final simulation time are presented in Fig. 3 as two-dimensional slices. The figures are centred on the mixing layer and use the same scale across all plots, showing different domains widths for the transverse strain cases and different mixing layer heights for the axial strain cases due to the effects of the strain.

Whilst the ILES cases are three-dimensional, the homogeneity of the y–z planes allows the RANS simulations to be conducted as one-dimensional, using a mesh of 720 cells in the x-direction. The domain is initialised using the planar-averages of the initial conditions for the quarter-scale θ-group case. The fluids are set up in a heavy-to-light configuration with densities of $3\,\mathrm{kg\,m^{-3}}$ and $1\,\mathrm{kg\,m^{-3}}$ to achieve a 0.5 Atwood number. The shocked heavy fluid is initialised for $x < 3.5\,\mathrm{m}$ using a $Ma = 1.8439$ shock for a pressure ratio of four. The interface between the two fluids is located at $x_0 = 3.5\,\mathrm{m}$ with a diffuse error function profile. As the mean profile is applied, the diffusive thickness of the initial interface is taken to use



(a) Axial compression (ACS-1)

(b) Axial expansion (ACS+1)

(c) Unstrained (S0)

(d) Transverse compression (TCS-1)

(e) Transverse expansion (TCS+1)

FIG. 3: Contours of the mass fraction of species 1 at $\tau = 34.5$ along the centre-plane from the ILES constant strain rate cases. Major ticks indicate a distance of 1 m.

the standard deviation of the ILES perturbation spectrum. The volume fraction profile used to initialise the mass fraction profile is

$$\bar{f}_1(x) = \frac{1}{2}\left(1 - \operatorname{erf}\left(\frac{x - x_0}{\sqrt{2}\sigma_0}\right)\right). \tag{30}$$

The standard deviation σ_0 is $0.1\lambda_{\min}$, where the initial perturbation wavelengths ranged between $2\pi/32$ and $2\pi/16$. The initial turbulent kinetic energy is set to a nominal value of $\tilde{K} = 10^{-40}\mathrm{m}^2\,\mathrm{s}^{-2}$ throughout the domain, whilst the initial turbulent length scale is calibrated to reproduce the integral width of the unstrained ILES case, i.e. the original quarter-scale θ-group simulation. The details of the turbulent length scale are further discussed in §II E. Both fluids use an ideal gas equation of state with $\gamma = 5/3$ and have an unshocked pressure of 100 kPa. The whole domain has an initial velocity offset of $u_1 = -291.575\,\mathrm{m\,s}^{-1}$ to counteract the velocity shift from the interface-shock interaction, making the post-shock interface stationary. At the non-dimensional time $\tau = 1$ the strain is applied to the simulation domain, just as was performed with the ILES cases.

The K-L model, like most RANS models, is not capable of effectively capturing the growth rate of the mixing layer at all stages of the instability evolution. Instead the focus during model calibration is on predicting the late-time growth rate for the mixing layers as opposed to the linear regime growth rate. By calibrating the initial conditions of the K-L model to match the late-time integral width for the unstrained case, the RANS results are not aligned during the linear or transitional regimes. Improved alignment of results in these regimes may be possible with the introduction of an intermittency factor which serves to quantify the local transition to turbulence and correspondingly limit the turbulent viscosity [75]. Using the K-L model coefficients posited in Refs. [48, 49], it is possible to capture the integral width of the unstrained case at late-time, however the turbulent kinetic energy results are smaller than the ILES. To provide a better comparison with the ILES data, the coefficients of the model were re-calculated so that both the integral width and turbulent kinetic energy results are aligned for the unstrained simulations at late-time. The coefficients used for the simulations in the present study are listed in Table II, which were derived using the same methodology laid out in the aforementioned articles. This includes using $\theta = 1/4$, which is lower than the value of $\theta = 0.291$ measured at late-time [74] or the value of $\theta = 1/3$ used in the buoyancy-drag model [76, 77]. The value of θ is time-dependent for the quarter-scale case, likely reaching a value of $\theta = 1/4$ for even longer simulation times. The calibrating coefficient of α, which scales the Rayleigh-Taylor growth, was adjusted from the default value of 0.05 to 0.01 to improve the turbulent kinetic energy calculation. This is a smaller value than $\alpha = 0.025$, which is the observed coefficient for narrowband RTI simulations. This adjustment allows the K-L model to simultaneously capture the mixing zone growth and the decay of the turbulence, however it causes the model to underestimate

TABLE II: Coefficients for the K-L turbulence model

Model	C_A	C_B	C_D	C_L	C_P	C_μ	N_H	N_K	N_L	N_Y
Refs. [48, 49]	11.2	0.76	0.2	0.19	2/3	1.19	0.35	0.43	0.04	0.35
Current	5.01	0.34	0.2	0.19	2/3	1.19	1.74	2.14	0.19	1.74

the Rayleigh-Taylor growth rate. The ability of the K-L model to capture the influence of the strain rate on the RMI-induced mixing layer is not affected by this recalibration, as simulations conducted with the original coefficients of Refs. [48, 49] demonstrated the same trends as observed in the results presented here-within. The choice of coefficients employed simply allow the turbulent kinetic energy comparison between the ILES and RANS results to be conducted more easily as the unstrained profiles are calibrated to be aligned. Using the original coefficients and matching the integral width causes the turbulent kinetic energy from the RANS to be around a factor of five smaller than the ILES results.

E. Bulk compression closure

In the prescribed model, there are two coefficients which are not determined from the self-similarity analysis. The first is C_P, which is the contribution of the turbulent kinetic energy to the Reynolds stress components. The second term is the coefficient for the bulk compression of the turbulent length scale, C_C. The value of this term is typically 1/3 under the assumption of conservation of mass in an eddy under isotropic compression [12]. If the mass inside the eddy is given by ρL^3, then for isotropic compression

$$\frac{\mathrm{d}L}{\mathrm{d}t} = \frac{1}{3} L \frac{\partial u_k}{\partial x_k}. \tag{31}$$

More generally, this approach will evolve the turbulent length scale according to the mean strain rate of the local flow field. In contrast, the multi-fluid model of Andrews [78] scales the length scale with the negative of the strain tensor magnitude, whilst the multi-fluid of Youngs [79] scales with the divergence but also considers the direction of the strain, using the strain in the direction of mixing as determined by the gradient of the species' volume or mass fraction. In the cases where the expansion/compression only occurs in n dimensions, Morel & Mansour [65] consider the conservation of ρL^n instead. The resulting evolution of the length scale is

$$\frac{\mathrm{d}L}{\mathrm{d}t} = \frac{1}{n} L \frac{\partial u_k}{\partial u_k}. \tag{32}$$

Under anisotropic strain rates, length scales in different directions will vary depending upon the strain rate that is aligned with the length scale. Considering the axial and transverse strain rates, each strain rate will modify the corresponding length scale. From this, two alternative closures for the bulk compression term are introduced, which use either the axial strain rate or the transverse strain rate:

$$\frac{\mathrm{d}L}{\mathrm{d}t} = L \frac{\partial u_1}{\partial x_1} = L S_A, \tag{33}$$

$$\frac{\mathrm{d}L}{\mathrm{d}t} = L \frac{1}{2} \left(\frac{\partial u_2}{\partial x_2} + \frac{\partial u_3}{\partial x_3} \right) = L S_T. \tag{34}$$

Using the approach of Mansel & Mansour [65], the axial closure would be obtained if only axial strain was expected, whilst the transverse closure would be obtained under transverse strain. For implementation purposes, the strain rates used in the bulk compression term are derived from the Favre-averaged strain rates, $\tilde{u}_i$, as these are the quantities readily available from the transport equation. An alternative approach could be to use the Reynolds-averaged velocity field which does not include the density weighting. This Reynolds-averaged velocity could be calculated more easily for a K-L-a model, using the calculation $\bar{u}_i = \tilde{u}_i - a_i$ for the turbulent mass flux a_i. The effects of the choice of formulation could be investigated, however this would be best performed with a case at a higher Atwood number to further exacerbate these effects.

These alternative closures for the bulk compression of the turbulent length scale can also be considered as using different C_C coefficients for the different cases. An equivalent C_C coefficient can be calculated for each type of strain application,

$$\hat{C}_C \frac{\partial u_k}{\partial x_k} = S_\phi \tag{35}$$

where S_ϕ represents the desired closure strain rate (S_A, S_T, or $S_I = \nabla \cdot u/3$), and $\hat{C}_C$ is the equivalent coefficient. These coefficients are given in Table III. For isotropic strain rates, all three closures will reduce to the same scaling as the mean strain rate is equivalent to the strain rate in any direction. Under pure axial strain, the transverse strain rate closure reduces $\hat{C}_C$ to zero as the bulk compression term will not activate, whilst the axial strain

TABLE III: The $\hat{C}_C$ coefficients for the different bulk compression closures under different applied strain rates, representing the equivalent C_C coefficient/scaling of the velocity divergence for the specified applied strain rates.

	Applied strain rate		
Length scale closure	Axial strain	Isotropic strain	Transverse strain
Axial, S_A	1	1/3	0
Isotropic, S_I	1/3	1/3	1/3
Transverse, S_T	0	1/3	1/2

closure increases the coefficient to unity. The equivalent coefficients assume that the term only activates under bulk compression/expansion, however the utilisation of Favre averaged velocities in the divergence means that it is affected by the mixing of fluids at different densities.

Given the possibility of the axial, isotropic or transverse strain closure, the simulation cases listed in Table I were conducted for each closure, providing three sets of RANS data. The effect of the different approaches manifests in the initialisation of the flow variables in the domain. A common approach to initialising the interface for RMI is to apply a non-negligible L at the cells adjacent to the interface and zero elsewhere, whilst $\tilde{K}$ is set to a negligible value everywhere. When the shock interacts with the interface, the buoyancy production term will activate, generating turbulent kinetic energy at the interface. The amount of turbulent kinetic energy deposited, and the resulting growth rate, will depend on the initial value of L. The bulk compression term will also activate due to the velocity gradient across the shock, and as a result different initial L values are required under the different scaling schemes, as shown in Table IV. As the shock is compressive and acts in the axial direction, the axial strain closure simulations require a larger initial L value to counteract the resulting bulk compression of L and achieve the same post-shock growth rate prior to the application of strain. The listed values of L are obtained by calibrating the initial L value in the cells on either side of the interface to predict the unstrained growth rate. Outside of the cells neighbouring the interface, the value of $L_0 = 1 \times 10^{-16}$ m is used.

TABLE IV: Values of L used for simulation initialisation

Axial closure	Isotropic closure	Transverse closure
0.54m	0.53m	0.52m

III. RESULTS AND DISCUSSION

A. Integral Width

The mixing layer width is measured here using the integral width, which is based upon the mean volume fraction profile,

$$W = \int \bar{f}_1 \bar{f}_2 dx. \tag{36}$$

The original ILES cases were conducted using the five-equation model of Allaire *et. al* [80] which includes a transport equation for the volume fraction of each species. As the K-L model only transports the mean mass fraction, the volume fraction is calculated assuming equal temperature and pressure within the cell, allowing the volume fraction of species 1 to be calculated by

$$\bar{f}_1 = \frac{\tilde{Y}_1/W_1}{\sum_a \tilde{Y}_a/W_a}. \tag{37}$$

The molecular mass, W_a, of each species is $90\,\mathrm{g\,mol^{-1}}$ and $30\,\mathrm{g\,mol^{-1}}$ respectively, inline with the initial density ratio between the fluids. The integral width results for the simulations with axial strain rates applied are shown in Fig. 4 for the three K-L models and the ILES results. The influence of the axial strain rate on the integral width demonstrates an increased integral width for expansive strain rates, as the velocity difference across the mixing layer causes the mixing layer to stretch/compress for positive/negative strain rates [81]. The K-L models are unable to capture the early-time growth of the mixing layer as the models are designed for late-time self-similar growth, and as a result are not well designed for the transitional regime. Despite the underestimation of the mixing layer growth rate at early time, the K-L models show the same increased growth rate under expansion strain as observed in the ILES. This stretching/compression effect arises from the velocity difference in the mean flow field which is resolved by the simulation, allowing the simulations to capture this direct effect of the axial strain rate. The variation between the different bulk compression closures arises from the turbulent growth, which is dependent upon the turbulent viscosity and proportional to L and $\tilde{K}^{1/2}$. The value of L is modified by the bulk compression term, with greater variation of L under strain for configurations with larger $\hat{C}_C$ values. This effect on the mixing layer is evident as the axial strain closure with the applied axial strain ($\hat{C}_C = 1$) will have the largest modification of L from the bulk compression term, and is observed to overestimate the influence of the strain rates on the mixing layer. The model with the best performance, most accurately capturing the turbulent growth, is the transverse strain closure implementation with $\hat{C}_C = 0$, effectively not modifying the turbulent length scale for the cases with only axial strain applied. The default closure with $\hat{C}_C = 1/3$ performs quite well for this problem, however the model can be observed to overestimate the influence of the integral width of the mixing layer. To compare the accuracy of the three implemented models, the mean absolute percentage error (MAPE) between the RANS results and the ILES data is recorded for the low-magnitude strain rate cases (cases ACV$\pm$1 and ACS$\pm$1) at the final time. The high-magnitude strain cases are omitted as their final results are still polluted by the underestimation of the transitional regime by the K-L model. The MAPE of the four cases are recorded in Table V, and shows the transverse closure to reduce the MAPE by a factor of four compared to the default isotropic closure, whilst the axial closure performs closer to 2.5$\times$ worse. The failings of the axial strain closure under the applied axial strain rates can be further discerned through the lens of self-similarity. The axial strain closure allows the turbulent length scale to grow in proportion to the mixing layer which is stretched/compressed, maintaining a self-similarity between the two, and is further discussed in §III E. This self-similarity is not representative of the actual flow, with the ILES results showing diverging evolutions for the turbulent kinetic anisotropy and mixedness. It is therefore unsurprising that a model that assumes self-similarity does not perform well when the flow is not actually self-similar.

Under applied transverse strain rates, the ILES obtains slightly larger growth rates for the expansive strain as compared to the unstrained simulation. In Ref. [20], this trend was considered the result of the turbulence length scale in the simulation stretching under expansion, which in turn effectively decreased the dissipation rate. A buoyancy-drag model calibrated to the unstrained case [77] was modified for the applied transverse strain cases by scaling the unstrained effective drag length scale by the transverse expansion factor,

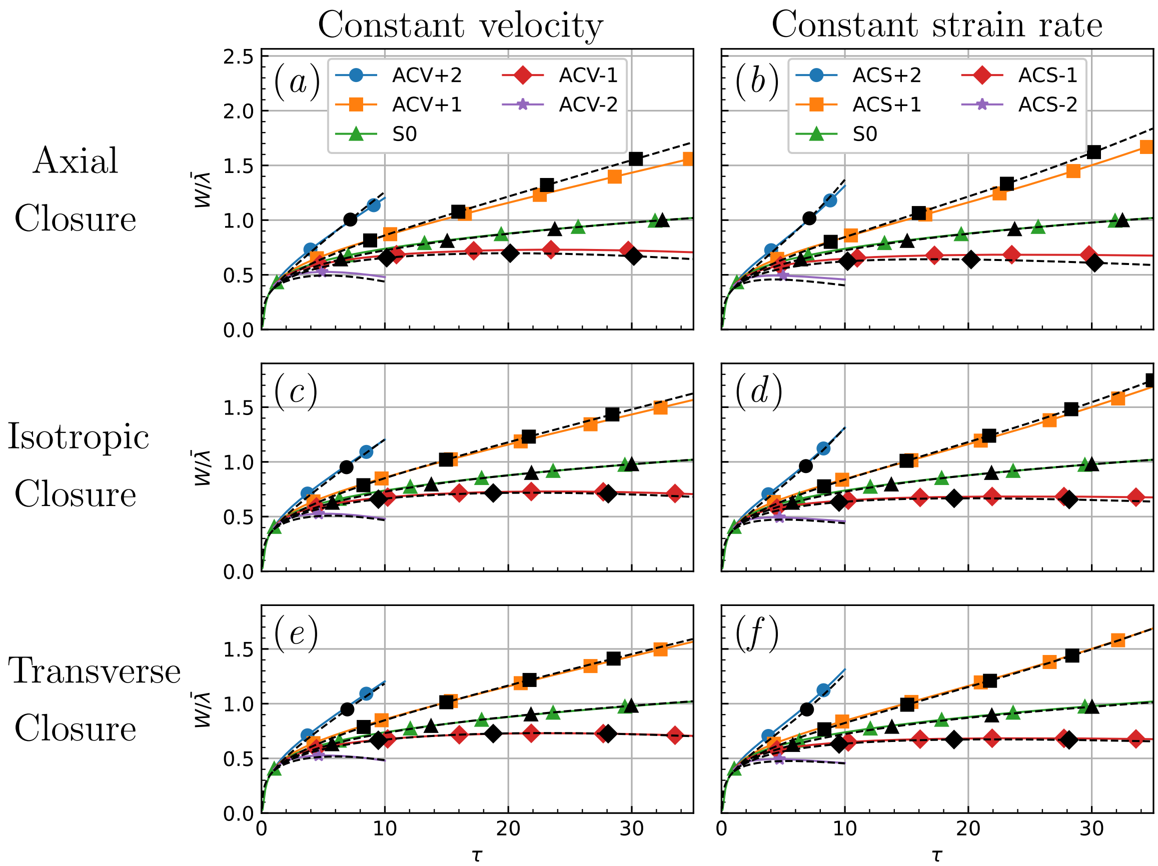


FIG. 4: Integral width for simulations under the applied axial strain rate with (left) constant velocity profile and (right) constant strain rate profile. Solid lines indicate ILES, dashed lines indicate K-L model with (top) axial closure, (middle) isotropic closure, and (bottom) transverse closure for bulk compression.

with the resulting model able to accurately predict the integral width [20]. For the K-L model simulations the integral width predictions vary, as shown in Fig. 5. Just as observed for the ILES, the K-L model results show only a small modification to the integral width, as the transverse strain does not directly stretch/compress the mixing layer as the axial strain does. The isotropic closure shows almost no variation of the integral width, as if the effects of shear production and bulk compression cancel each other out. Under compression, shear production will deposit turbulent kinetic energy and the bulk compression will reduce the turbulent length scale, which can effectively maintain a steady turbulent viscosity and growth rate. Using the axial closure, corresponding to $\hat{C}_C = 0$, causes the K-L model to incorrectly predict the sign of the change, with the expansion cases growing at a slower

TABLE V: Mean absolute percentage error (MAPE) at the final simulation time for the low-magnitude strain cases. Closure model with lowest MAPE for each strain application are highlighted in bold text.

		MAPE	
Strain Direction	Closure	Integral Width	TKE
Axial	Axial	9.91	68.9
Axial	Isotropic	4.12	39.9
Axial	**Transverse**	**1.28**	**25.9**
Transverse	Axial	9.55	47.7
Transverse	Isotropic	3.77	20.4
Transverse	**Transverse**	**0.80**	**7.2**

rate than the unstrained case. As there is no bulk compression for this closure and strain combination, the only effect would be shear production, which would decrease the turbulent kinetic energy and turbulent viscosity in turn, producing the observed results for expansive strain. Using the transverse strain closure improves the performance of the model, aligning to the ILES data more effectively than the isotropic or axial closure. The transverse closure has a larger effective C_C ($\hat{C}_C = 1/2$), causing the turbulent length scale to vary more under the transverse strain. The MAPE for the low-magnitude strain cases is listed in Table V, and the results confirm that the transverse closure is the most well aligned of the three closure methods. Similar ratios are observed as for the applied axial strain cases, with the transverse closure of the turbulent length scale attaining a factor of four smaller MAPE than the isotropic closure.

A comparison of the volume fraction profile through the mixing layer is presented in Fig. 6 for the constant strain rate cases at the final time. The different profiles for the RANS simulations appear identical for the unstrained case. The strained cases differ, as expected from the integral width measurements, which is most obvious around the edges of the mixing layer where the difference in the mixing layer width is most obvious. For cases ACS+1 and TCS-1, the transverse closure has the most similar profile to the ILES, whilst the isotropic and axial closures estimate the mean profile to be consecutively wider. Likewise for cases

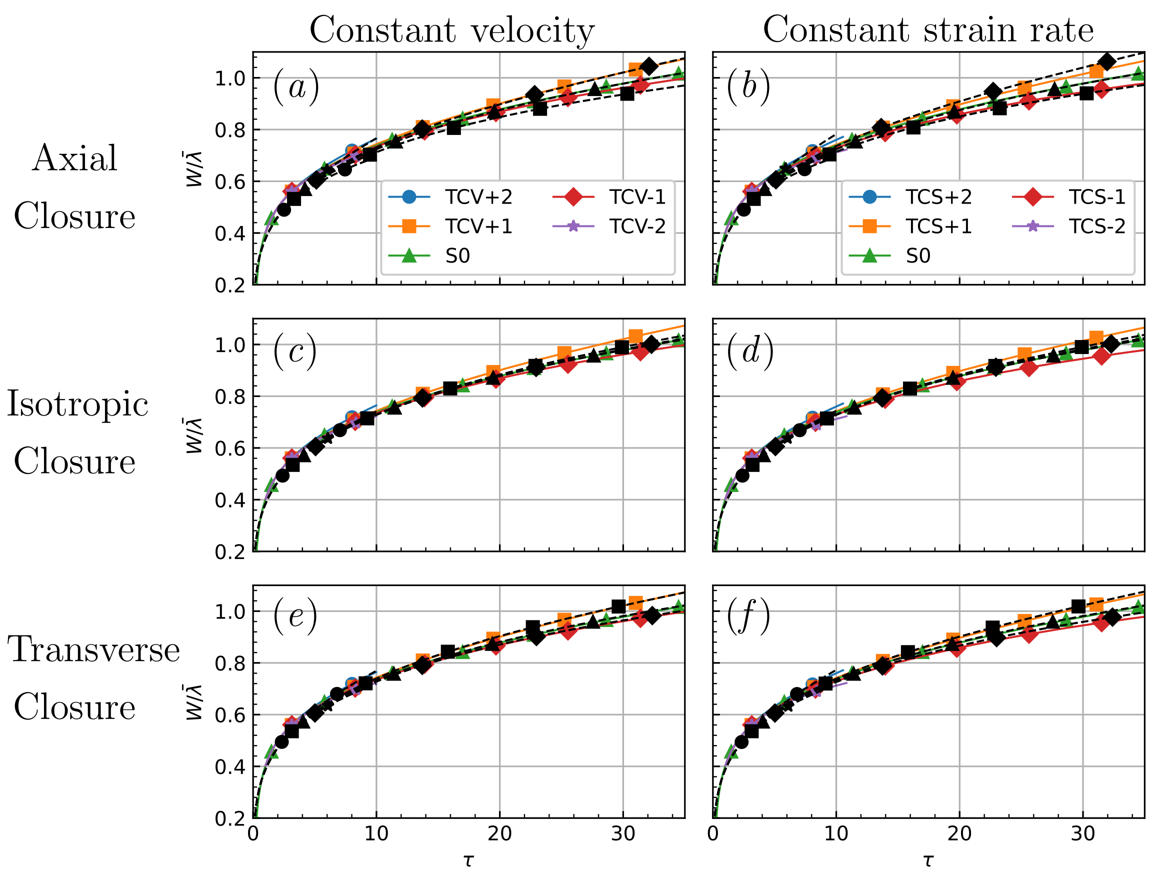


FIG. 5: Integral width for simulations under the applied transverse strain rate with (left) constant velocity profile and (right) constant strain rate profile. Solid lines indicate ILES, dashed lines indicate K-L model with (top) axial closure, (middle) isotropic closure, and (bottom) transverse closure for bulk compression.

ACS-1 and TCS+1, the axial and isotropic closures predict a slightly narrower mixing layer width.

B. Turbulent kinetic energy

The domain integrated turbulent kinetic energy is calculated by different methods for the ILES and RANS cases. For the ILES, cases the turbulent kinetic energy is evaluated using the Favre velocity fluctuations:

$$TKE = \iiint \frac{1}{2} \rho u_i'' u_i'' dV. \tag{38}$$

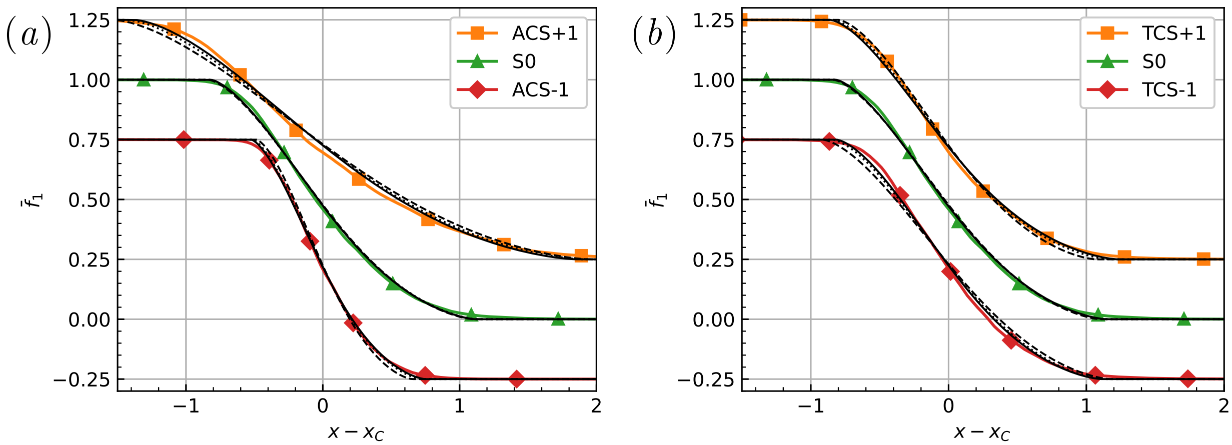


FIG. 6: Mean volume fraction profile at $\tau = 34.45$ for cases under (a) axial strain, and (b) transverse strain. Cases are displayed with a vertical offset of $\Delta \bar{f}_1 = 0.25$. Solid coloured lines represent ILES, solid black lines represent transverse closure, dotted black lines represent isotropic closure, and dashed black lines represent axial closure.

The RANS cases calculate the total turbulent kinetic energy by integrating the transported turbulent kinetic energy variable:

$$TKE = \iiint \bar{\rho}\tilde{K} dV. \tag{39}$$

The results for the axially strained cases are plotted in Fig. 7. The ILES results show the impact of shear production, with the compressive strain cases obtaining a slight increase in TKE, whilst the compression cases decrease. The axial closure RANS cases show little variation in the TKE, which can be explained by the change in the turbulent length scale affecting the dissipation rate and counteracting shear production. In contrast, the transverse closure cases do a better job of capturing the effect of the strain rate on the TKE. The MAPE for low-magnitude strain cases are calculated in the same manner as was done for the integral width, and is reported in Table V. The results show the same trend as for the integral width, with the transverse closure K-L model performing the best, and the axial closure model performing the worst. This is not surprising, as for the transverse closure to accurately capture the turbulent growth of the integral width, it would require more accurate representation of the turbulent viscosity and turbulent kinetic energy compared to the other methods. It is interesting to note that the transverse closure does a better job of capturing the TKE for the compressive cases, as compared to the expansion cases. As a two-equation

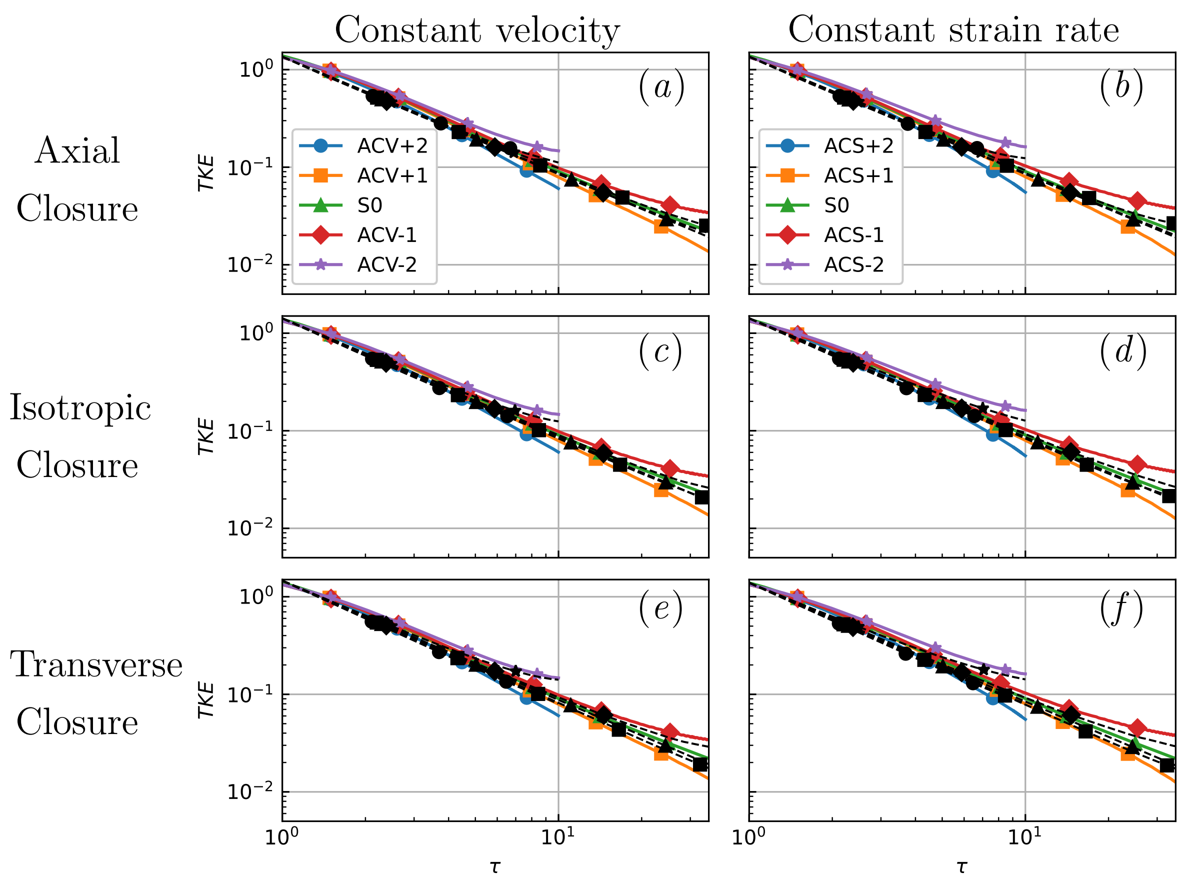


FIG. 7: Turbulent kinetic energy for simulations under the applied axial strain rate with (left) constant velocity profile and (right) constant strain rate profile. Solid lines indicate ILES, dashed lines indicate K-L model with (top) axial closure, (middle) isotropic closure, and (bottom) transverse closure for bulk compression.

model, the turbulent kinetic energy is assumed to be isotropic and is evenly divided across the three normal Reynolds stresses, as indicated by the $-C_P\bar{\rho}\tilde{K}\delta_{ij}$ contribution to $\bar{\tau}_{ij}$ in Eq. 16. The expansion cases tend towards isotropy for the TKE as the simulation continues, whilst the compression cases are anisotropic, with the TKE becoming more aligned with the x-direction [19]. It is interesting to note that the model performs better for these anisotropic cases than it does for the isotropic ones.

The domain integrated turbulent kinetic energy for the transverse strain cases is shown in Fig. 8. The ILES results show little variation as a result of the change in the turbulent length scale. A simple Reynolds stress model was calibrated to the unstrained case in Ref. [20], which showed that the TKE could be accurately estimated if the length scale for the

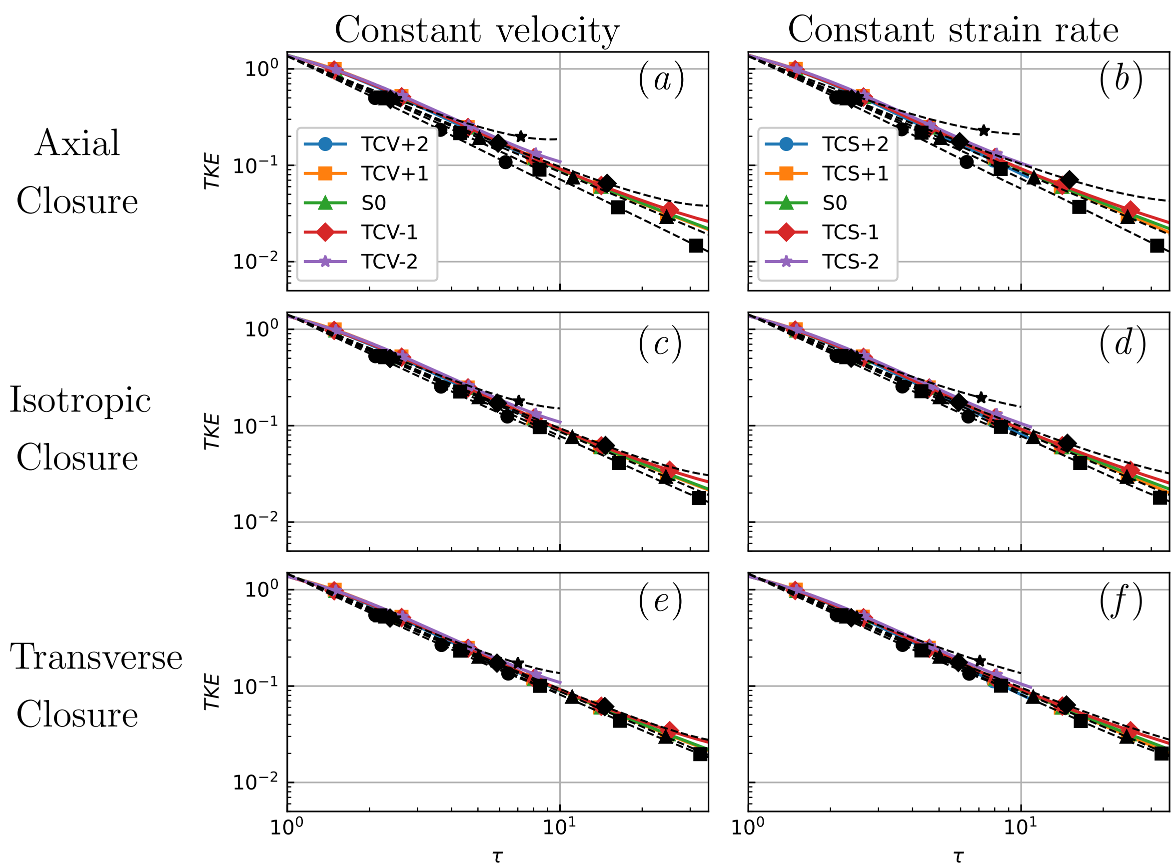


FIG. 8: Turbulent kinetic energy for simulations under the applied transverse strain rate with (left) constant velocity profile and (right) constant strain rate profile. Solid lines indicate ILES, dashed lines indicate K-L model with (top) axial closure, (middle) isotropic closure, and (bottom) transverse closure for bulk compression.

TKE dissipation was made to scale with the geometric mean of the expansion factor in each direction. For compression, this means the length scale decreases and in turn increases the dissipation rate, counteracting shear production that adds energy into the mixing layer from the negative velocity gradients. The counteracting behaviour causes the tight spread observed in the ILES results. Again, the transverse closure appears to do the best job of capturing the tight spread, with the isotropic closure showing a slightly larger variation, and the axial closure overestimating the influence of the strain rate. This is not surprising then, as the axial closure will not modify the turbulent length scale directly ($\hat{C}_C = 0$), so there is no counteracting behaviour to shear production. The MAPE is reported in Table V, with the transverse closure attaining the lowest error out of the three models.

The mean profiles for the turbulent kinetic energy are shown in Fig. 9 for the constant strain rate cases at the final time. For the unstrained case, the RANS simulations are identical and are able to reasonably predict the peak of the turbulent kinetic energy distribution. The RANS simulations, for all cases presented, are not able to predict the spike side turbulent kinetic energy, as the turbulent kinetic energy distribution is confined to the region in which the two fluids mix by the nature of the model. As the spike side is the region with the lowest density, the impact of this discrepancy is diminished by using $\bar{\rho}\tilde{K}$ for domain integration. The improvement of the transverse closure is most visible in cases ACS+1 and TCS-1, where the axial and isotropic closures overestimate the increase of turbulent kinetic energy. For cases ACS-1 and TCS+1, the isotropic closure has better alignment along the bubble side of the profile, however the transverse closure's slight overestimation allows for a better estimate of the domain integrated turbulent kinetic energy, as is done for the unstrained case.

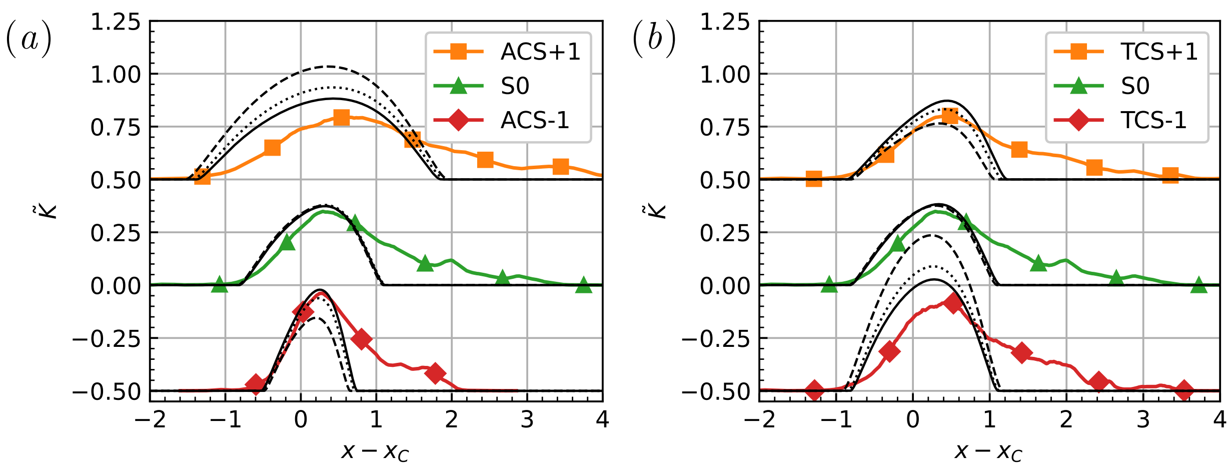


FIG. 9: Mean turbulent kinetic energy profile at $\tau = 34.45$ for constant strain rate cases under (a) axial strain, and (b) transverse strain. Cases are displayed with a vertical offset of $\Delta\tilde{K} = 0.50$. Solid coloured lines represent ILES, solid black lines represent transverse closure, dotted black lines represent isotropic closure, and dashed black lines represent axial closure.

C. Turbulent mass flux velocity

The addition of a transport equation for the turbulent mass flux velocity, a_i, is the most common way to improve upon the two-equation models for compressible flows [42, 53, 59]. With the addition of the turbulent mass flux velocity, the system is able to more accurately treat the conversion of potential energy to turbulent kinetic energy and avoid the shock detection mechanism outlined in equation (17). After the shock transit, the influence of the turbulent mass flux is negligible for the developing turbulent mixing layer, as it only modifies the values of L and $\tilde{K}$ due to buoyancy production when $\partial \bar{p}/\partial x_i \neq 0$. The turbulent mass flux is calculated according to,

$$a_i = \tilde{u}_i - \bar{u}_i = \overline{-u_i''}. \tag{40}$$

Whilst the implemented K-L model does not calculate a_i, it is possible to estimate the value for a_i for incompressible flows [82]. At a sufficiently late time, the decaying turbulent mixing layer induced by RMI may become approximately incompressible. The value for a_1 can be approximated by using the Favre-weighted velocity from the simulation for $\tilde{u}_1$, and using $\bar{u}_1 = \bar{S}_A$. A comparison of the ILES results and the estimates from each closure is presented in Fig. 10 for the constant strain rate cases near the final time. The RANS results for the unstrained case are identical for all closures, showing agreement with the ILES results but not fully capturing the spike side profile. The transverse closure results provide the best overall estimate of the profile for the strained cases. The ordering and effect of the closure on the turbulent mass flux velocity is the same as observed for the turbulent kinetic energy. For this reason, the transverse closure is expected to be beneficial for models which also transport the turbulent mass flux velocity, utilising the modified turbulent length scale for improved dissipation estimates.

D. Reynolds Stress

One of the weaknesses of the K-L model is the inability to estimate the Reynolds-stresses, relying on the Boussinesq eddy viscosity assumption. In the presence of strong velocity gradients, as observed with shock waves, the Reynolds stress tensor can become non-realisable. There are different approaches to avoid this, such as only using the isotropic turbulent pressure component [12], limiting the obtained stresses [82], or modifying the eddy viscosity to

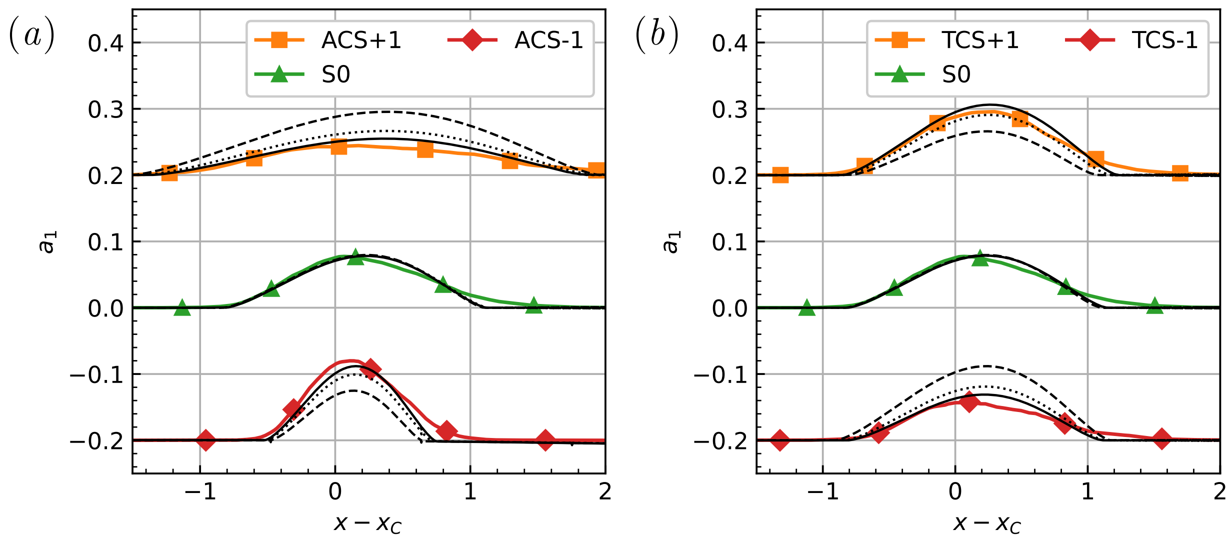


FIG. 10: Turbulent mass flux velocity profile at $\tau = 34.45$ for cases under (a) axial strain, and (b) transverse strain. Cases are displayed with a vertical offset of $\Delta a_1 = 0.20$. Solid coloured lines represent ILES, solid black lines represent transverse closure, dotted black lines represent isotropic closure, and dashed black lines represent axial closure.

ensure realisability [52]. For the simulations conducted, negligible differences were obtained for the integral width and total turbulent kinetic energy whether using the limited or unlimited Reynolds stress components. The Reynolds stress components are calculated using the Boussinesq eddy viscosity assumption for the RANS cases, according to

$$\widetilde{u_i u_j} = -\frac{\bar{\tau}_{ij}}{\bar{\rho}} = \frac{2}{3}\tilde{K}\delta_{ij} - \frac{\mu_T}{\bar{\rho}}\left(\frac{\partial \tilde{u}_i}{\partial x_j} + \frac{\partial \tilde{u}_j}{\partial x_i} - \frac{2}{3}\frac{\partial \tilde{u}_k}{\partial x_k}\delta_{ij}\right). \tag{41}$$

The axial $(\widetilde{u_1''u_1''})$ and transverse $(\widetilde{u_2''u_2''})$ stresses at the final time for the constant strain rate case are shown for the axially strained simulations in Fig. 11 and in Fig. 12 for the transversely strained simulations. The anisotropy of the Reynolds stress components is visible in the ILES results; however, the RANS cases do not perform well at correctly predicting the ratio. This is most evident for the $\widetilde{u_1''u_1''}$ component for case ACS+1 which becomes non-realisable in the absence of any correction. This is accompanied by an over-estimate of the transverse stress, however using the transverse closure minimises the under- and over-prediction of the stresses. Similar behaviour is observed for case TCS-1 which has non-realisable $\widetilde{u_1''u_1''}$ values at edges of the captured mixing layer. Part of the difficulty in the capturing the cases ACS+1 and TCS-1 is that near the final time the turbulent kinetic energy is close to isotropic, which is visible in the ILES profiles. As the unstrained case starts with the anisotropic distribution, the application of the strain rates causes these two cases to become less anisotropic, which the Boussinesq eddy viscosity assumption does not predict.

The isotropic and axial closures allow for better estimates of $\widetilde{u_1''u_1''}$ for case ACS-1 in Fig. 11, where as the transverse closure overestimates the stress. The other closures will then more accurately predict the shear production contribution. However, in the actual $\tilde{K}$ profile in Fig. 9, the profile is underestimated, a result of the closure which means the axial compression reducing the length scale and increasing the dissipation. In contrast, the transverse closure is consistent in the slight over-prediction of the turbulent kinetic energy.

Modelling the flow with a Reynolds stress model should improve the results due to the more accurate calculation of the shear production. However, the return to isotropy treatment of many models may not be well suited for the case considered, where the base case remains anisotropic at late time.

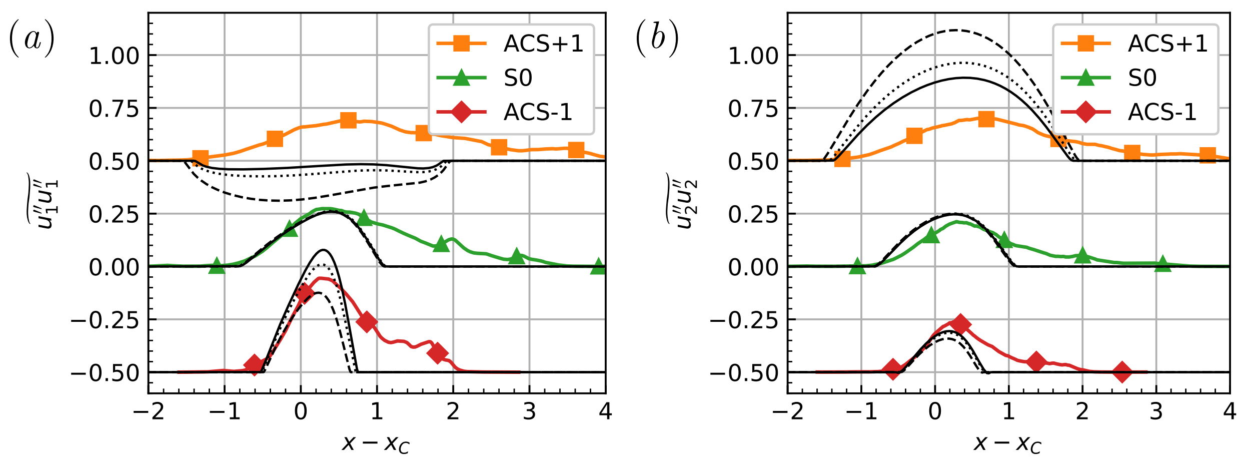


FIG. 11: Reynolds stress components at $\tau = 34.45$ for cases with constant axial strain rates. (a) $\widetilde{u_1''u_1''}$, and (b) $\widetilde{u_2''u_2''}$. Cases are displayed with a vertical offset of $\Delta\widetilde{u''u''} = 0.50$. Solid coloured lines represent ILES, solid black lines represent transverse closure, dotted black lines represent isotropic closure, and dashed black lines represent axial closure.

E. Self-similar analysis of the K-L model

In order to better understand the K-L model's performance, it is possible to perform a self-similarity analysis of the governing equations. This will reduce the equations to the

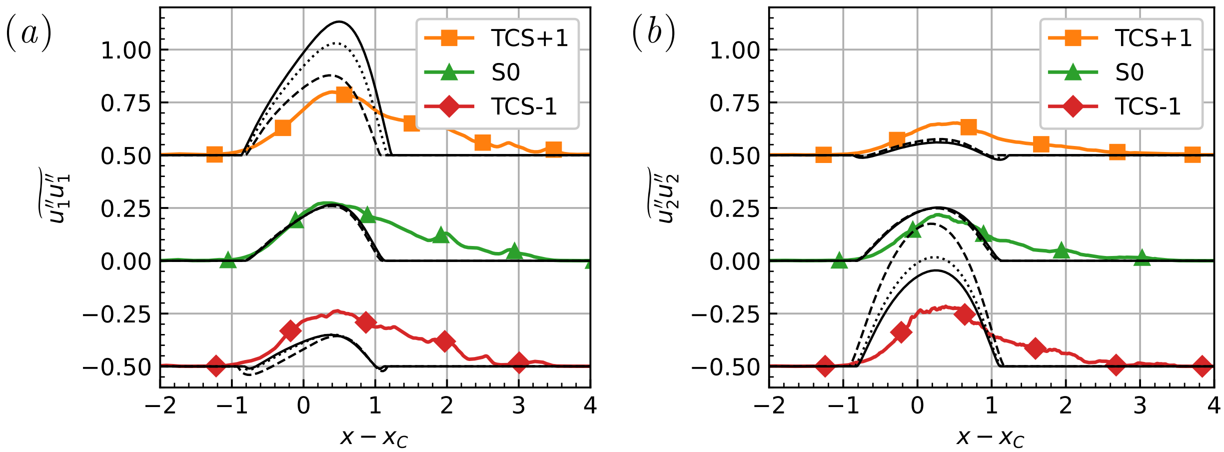


FIG. 12: Reynolds stress components at $\tau = 34.45$ for cases with constant transverse strain rates. (a) $\widetilde{u_1''u_1''}$, and (b) $\widetilde{u_2''u_2''}$. Cases are displayed with a vertical offset of $\Delta\widetilde{u''u''} = 0.50$. Solid coloured lines represent ILES, solid black lines represent transverse closure, dotted black lines represent isotropic closure, and dashed black lines represent axial closure.

form of a buoyancy-drag model, a set of ordinary-differential equations for the evolution of the mixing layer width and mixing layer growth rate. Following the work in Refs. [47–49], the profiles of L and $\tilde{K}$ are given by self-similar profiles:

$$L(x,t) = L_0(t)X^r(x,t) \tag{42}$$

$$\tilde{K}(x,t) = \tilde{K}_0(t)X^s(x,t) \tag{43}$$

where the profiles are effectively separated into a magnitude component, and a spatial component using the transformed coordinates:

$$X(x,t) = 1 - \chi^2 \tag{44}$$

$$\chi(x,t) \quad = \tfrac{x}{h(t)} \tag{45}$$

where h is half the mixing layer width. The non-dimensionalised spatial variables are limited to within the mixing layer i.e. $\chi \in [-1,1]$ and $X \in [0,1]$. Previous works assumed the values for $r = 1/2$ and $s = 1$ such as in Ref. [12] for the K-L model and Ref. [53] for the K-L-a model. Instead, Refs. [47–49] consider arbitrary values of s and r, with the constraint of $2r + s = 2$, in order to find the optimum fit to the models, ultimately using $s = 1.6$. The generalisation of the exponents meant that the self-similar equations did not separate easily into temporal and spatial contributions, and were solved by integrating over the mixing layer.

For the cases considered, the $\tilde{K}$ and L equations used will assume homogeneity in the transverse directions, and neglect the effects of density by assuming the flow is in the low-Atwood number limit.

$$\bar{\rho}\frac{D\tilde{K}}{Dt} = \frac{\partial}{\partial x_1}\frac{\mu_T}{N_K}\frac{\partial \tilde{K}}{\partial x_1} + \bar{\tau}_{ij}\frac{\partial \tilde{u}_i}{\partial x_j} - C_D\tilde{V}^3/L \tag{46a}$$

$$\bar{\rho}\frac{DL}{Dt} = \frac{\partial}{\partial x_1}\frac{\mu_T}{N_L}\frac{\partial L}{\partial x_1} + C_L\bar{\rho}\tilde{V} + \bar{\rho}LS_\phi \tag{46b}$$

The buoyancy production term for $\tilde{K}$ has been neglected, whilst the Reynolds stress contribution and the L compressibility term are included, two terms which are commonly ignored. For generality, the L bulk compression term has been represented as scaling with the arbitrary strain rate S_ϕ, where using the isotropic strain rate closure ($\nabla \cdot \boldsymbol{u}/3$) will provide the original model. To obtain a buoyancy-drag model, the self-similar profiles given in Eqs. (42) and (43) were substituted in, a Taylor series about $\chi = 0$ was taken to remove the spatial component, and the turbulent kinetic energy was transformed into the turbulent velocity using the relation $\tilde{K}_0 = \tilde{V}_0^2/2$. Assuming only normal velocity gradients are present, the gradients are replaced with the corresponding strain rate.

$$\frac{dL_0}{dt} = \tilde{V}_0(t)\left(\frac{C_L N_L - 2C_\mu r\beta^2}{N_L}\right) + L_0 S_\phi \tag{47a}$$

$$\frac{d\tilde{V}_0}{dt} = -\frac{\tilde{V}_0^2(t)}{L_0(t)}\left(\frac{C_D N_K + C_\mu s\beta^2}{N_K}\right) + \frac{4}{3}C_\mu L_0(S_A - S_T)^2 - \frac{1}{2}C_P(S_A + 2S_T)\tilde{V} \tag{47b}$$

Typically, the evolution equation of L_0 will be transformed into the evolution equation of the mixing layer half width, h, as these terms are considered to be proportional by a constant within the regime of self-similarity.

$$\beta = \frac{L_0(t)}{h(t)} \tag{48}$$

As the integral width has been used as the measure of the mixing layer width in the previous sections, the system will be transformed by $\beta_W = L_0/W$. However, the assumption of self-similarity is not valid for the strained simulation cases. Under the application of axial and transverse strain, the turbulent kinetic energy anisotropy and the mixedness of the mixing layer do not approach an asymptotic value, instead deviating away from the unstrained case's self-similar behaviour [19, 20]. This is further evidenced by the variation of β_W for the RANS

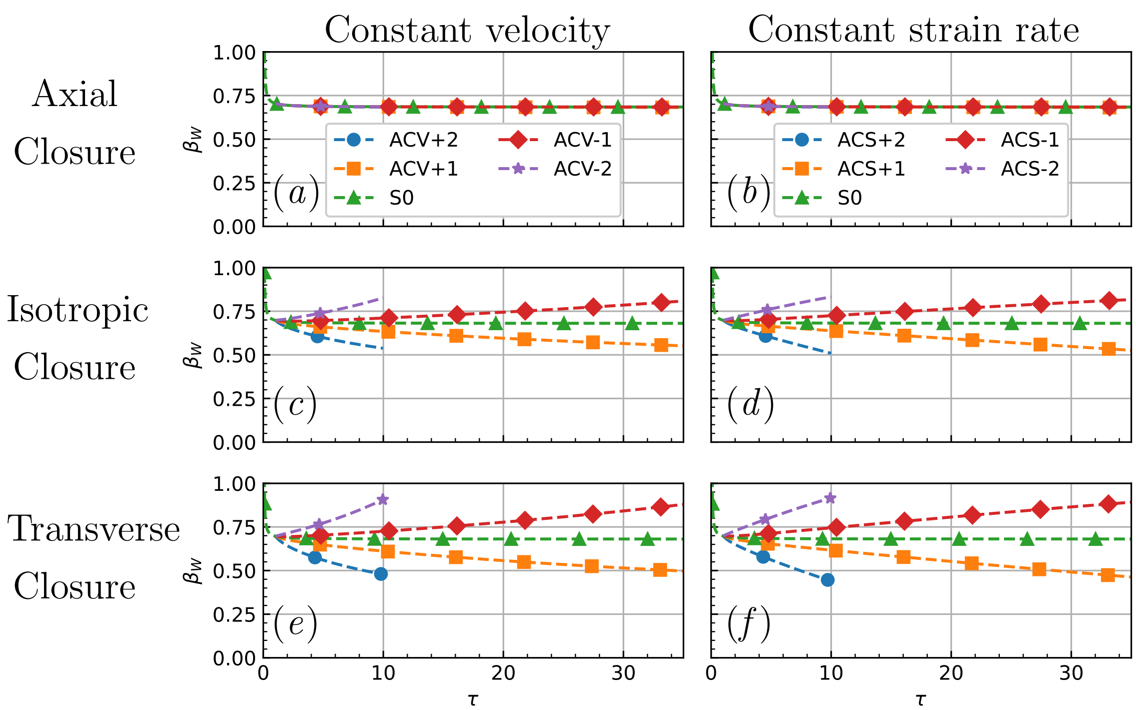


FIG. 13: β_W for simulations under the applied axial strain rate with (left) constant velocity profile and (right) constant strain rate profile for the K-L model with (top) axial closure, (middle) isotropic closure, and (bottom) transverse closure for bulk compression.

simulations conducted, shown in Figs. 13 and 14. The unstrained cases approach a final value of $\beta_W = 0.687$, suggesting the unstrained case has entered this regime of self-similar growth for the RANS model. It is possible to further validate this asymptotic value, as for a linear mean volume fraction profile the integral width should be three times smaller than the half width, giving the relation of $\beta_W = 3\beta$. The theoretical value of β for the present model is calculated according to $\beta = 1/C_A$ [47–49]. The actual relation observed from the simulations is $\beta_W = 3.44\beta$, a slightly higher ratio due to the mean volume fraction profile not being linear. Whilst the unstrained cases plateau, the strained cases show a variation in β_W, suggesting that the peak value of L in the domain and the integral width do not grow in the same proportion for the isotropic and transverse closure cases. In contrast, the axial closure cases do maintain the late-time value of β_W. Whilst the self-similarity assumption simplifies the buoyancy-drag reduction analysis, it is not an accurate representation of the state of self-similarity for the ILES cases. This necessitates the inclusion of time dependence for β.

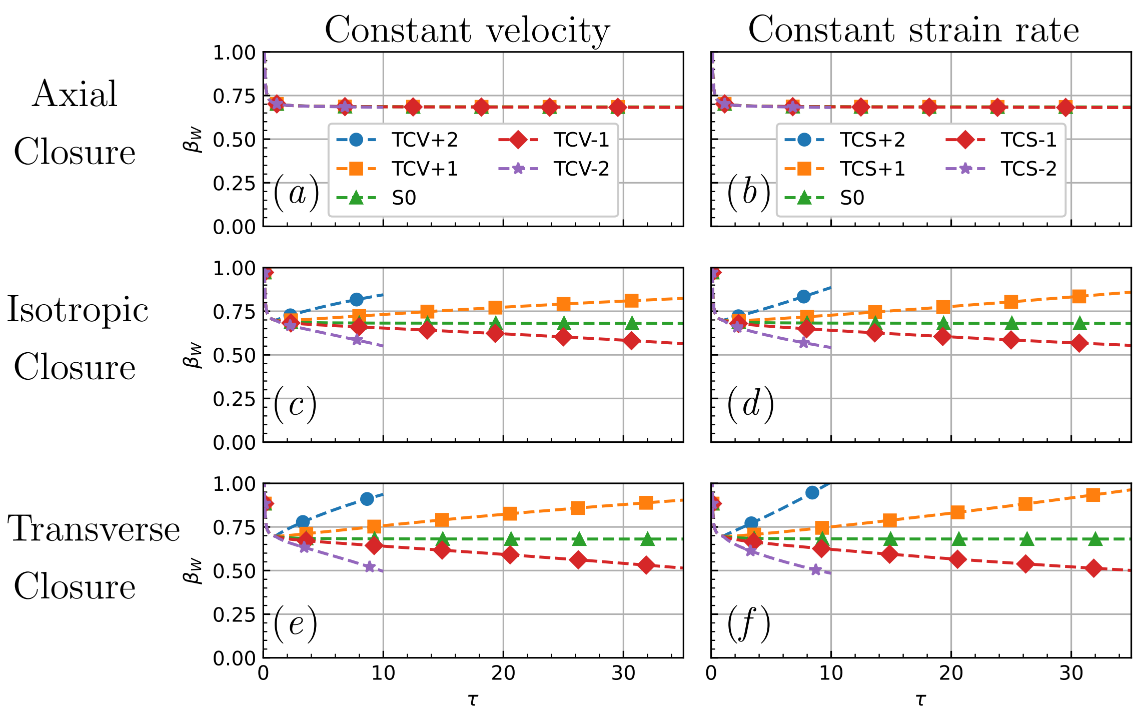


FIG. 14: β_W for simulations under the applied transverse strain rate with (left) constant velocity profile and (right) constant strain rate profile for the K-L model with (top) axial closure, (middle) isotropic closure, and (bottom) transverse closure for bulk compression.

Adjusting both equations to use the integral width, the growth rate of the integral width is defined by

$$V_W = \tilde{V}_0 \frac{C_L N_L - 2C_\mu r \beta^2}{N_L \beta_W}. \tag{49}$$

Substituting and simplifying gives the new expressions:

$$\frac{dW}{dt} = V_W + \left(S_\phi - \frac{\dot{\beta}_W}{\beta_W} \right) W \tag{50a}$$

$$\frac{dV_W}{dt} = -\frac{V_W^2}{l^{\text{eff}}} + C_W (S - S_T)^2 W - \frac{1}{2} C_P (S_A + 2S_T) V_W + C_\beta V_W \tag{50b}$$

with the definitions

$$l^{\text{eff}} = \frac{N_K \left(C_L N_L - 2C_\mu r \beta^2 \right)}{N_L \left(C_D N_K + C_\mu s \beta^2 \right)} W \tag{51a}$$

$$C_W = \frac{4C_\mu (C_L N_L - 2C_\mu r \beta^2)}{3N_L} \tag{51b}$$

$$C_\beta = -\frac{\dot{\beta}_W}{\beta_W} - \frac{4C_\mu r \beta}{C_L N_L - 2C_\mu r \beta^2} \dot{\beta}. \tag{51c}$$

This derivation has not assumed a constant β or β_W, causing the derivatives of each to occur in the differential equations. The equations may be simplified to use only a single β term as they are proportional for a specified mean volume fraction profile. The model is greatly simplified by using constant β. It may be possible to empirically determine a relationship for the variation in β under strain for the specific bulk compression closure employed. For now, the analysis is conducted with the assumption of self-similarity between the turbulent length scale and the mixing layer width. Using this assumption reduces the model to,

$$\frac{dW}{dt} = V_W + S_\phi W \tag{52a}$$

$$\frac{dV_W}{dt} = -\frac{V_W^2}{l^{\mathrm{eff}}} + C_W(S_A - S_T)^2 W - \frac{1}{2}C_P(S_A + 2S_T)V_W. \tag{52b}$$

In this model the growth of the integral width is dependent upon the velocity as well as a contribution from the background strain rate, $S_\phi W$. For the case of axial closure ($S_\phi = S_A$), this contribution is identical to the RMI linear regime correction with axial strain, where the amplitude growth rate is the superposition of the amplitude growth rate and the background stretching contribution [18, 19]. It makes sense then that axial closure preserves the β_W ratio, as it allows the turbulent length scale to grow at the same rate as the integral width, where the integral width will naturally vary from the velocity difference across the mixing layer. Despite this, the transverse closure provides better performance for the model by adjusting the dissipation rate for the transverse strain expansion factor. Without any strain and with the coefficients provided in Table II, the effective drag lengthscale is calculated to be $l^{\mathrm{eff}} = 0.383W$, which for unstrained RMI suggests a power law growth $W = A(t-t_0)^\theta$, where $\theta = 0.277$. This is in between the prescribed coefficient used for the K-L model derivation of $\theta = 0.25$, and the late-time coefficient of $\theta = 0.292$ for the quarter-scale θ-group case [74].

The velocity equation has two additional terms that are dependent upon strain rate. These two terms are the result of shear production, a product of the Reynolds stress and velocity gradients. Similar terms were hypothesised and calibrated to improve the performance of a buoyancy-drag model describing the ILES cases under axial strain [19]. Whilst the C_W term will always act as a positive source for the evolution of the growth rate, the C_P contribution will depend upon the signs of the applied strain rate as expected for shear production contribution.

Given the clear separation in behaviour of the two length scales, the integral width which scales with axial strain and the turbulent length scale which scales with transverse strain, a two length scale approach could be used to generate a buoyancy-drag model. Whilst two length scales are employed in many modern RANS models, those length scales are used to separate the turbulence transport and destruction length scales [55, 61]. In the derivation of the K-2L-a model, the turbulent length scale equation is considered to grow self-similarly with the mixing layer width, while the ratio between the turbulent transport and destruction length scale is not necessarily constant [55].

Two further modifications have been made to obtain the buoyancy-drag model in Eq. 53 from Eq. 52. Firstly, the transverse strain contributions from shear production have been omitted from the evolution of the turbulent velocity growth to improve agreement with the transverse results. Whilst turbulent kinetic energy is added under compressive transverse strain, it is not distributed to the axial direction which the isotropic model for $\tilde{K}$ does not account for and so causes the mixing layer growth rate to be incorrectly estimated. An alternate modification would be to scale the turbulent velocity more strongly with the transverse strain rate to counteract this effect. Secondly, the magnitude of the coefficient of the $S_A V_W$ term has been increased to improve the alignment with the axial results, effectively increasing the strain drag. Whilst these modifications are phenomenological, they can be considered corrections to the prior removal of the terms which depended upon the time variation of β.

$$\frac{dW}{dt} = V_W + S_A W \tag{53a}$$

$$\frac{dL}{dt} = V_W + S_T W \tag{53b}$$

$$\frac{dV_W}{dt} = -\frac{V_W^2}{l^{\text{eff}}} + 0.143 S_A^2 L - 0.933 S_A V_W \tag{53c}$$

$$l^{\text{eff}} = 0.383 L \tag{53d}$$

For simplicity, both length scales are scaled by the turbulent velocity as opposed to being different by a factor of β_W. For a case without applied strain, the proposed model will reduce down to the more typical two-equation model for buoyancy-drag modelling. The performance of the model is plotted in Figs. 15 and 16, using initial conditions of [W_0,L_0,V_{W0}]=[0.00638, 0.00638,1679] and an initial time offset of $\tau = 0.08$ to better align with the shock transition. The initial length scale values use the post-shock integral width [74, 77], as with the omission of the source term in the analysis the resulting buoyancy-drag model is not able to capture the shock transition. As the model uses an effective drag length

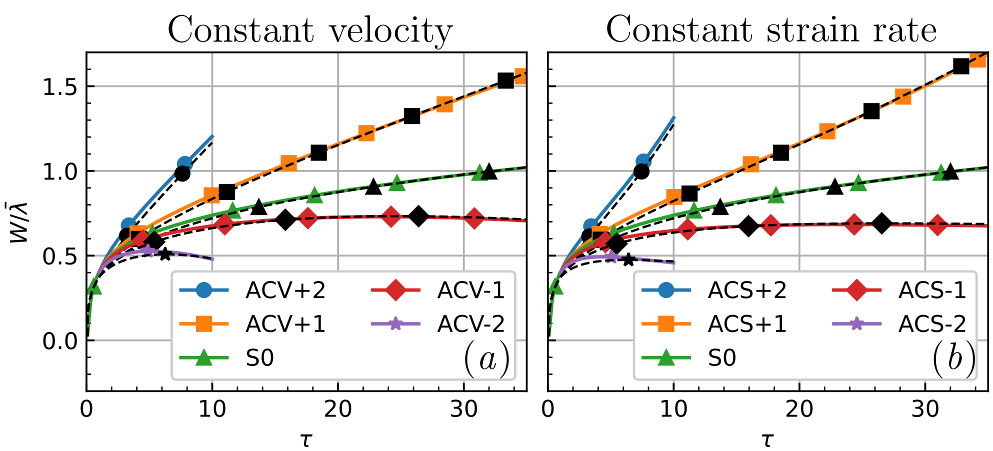


FIG. 15: Buoyancy-drag model for simulations under the applied axial strain rate with (left) constant velocity profile and (right) constant strain rate. Sold line indicates ILES, dashed line indicates buoyancy-drag model from Eq. 53.

scale proportional to the turbulent length scale which is initialised the same as the integral width, it is unable to capture the linear and transitional regimes, just like the RANS model. The initial drag is very high due to the small turbulent length scale/integral width, requiring a very large initial velocity (36× larger than the theoretical growth rate[74, 77]) to counteract the drag. This could be rectified by using a piecewise effective drag length scale as done in the buoyancy-drag models of Youngs and Thornber [76, 77]. As the drag term is derived from the diffusion and dissipation terms for the turbulent kinetic energy, a modification to these terms could improve the performance of the K-L model at early time. The late time performance of the derived buoyancy-drag model is well aligned with the ILES, and is far less computationally expensive to run than both the ILES and RANS simulations.

The buoyancy-drag model is viable for modelling RTI by including a buoyancy term of the form $C_b Atg$ in equation (53c) using the acceleration g. Whilst the coefficients of the RANS model were calibrated using the value of $\alpha = 0.01$ for RTI, the buoyancy-drag model growth can be calibrated through tuning the coefficient C_b in the buoyancy term. Assuming the late-time solution of the form $W \propto \alpha Atgt^2$ for unstrained RTI, the coefficients are related by the equation

$$C_b = 2\alpha \left(1 + \frac{2}{l^{\text{eff}}/W}\right). \tag{54}$$

For unstrained RTI, the effective drag length scale and the width are related by $l^{\text{eff}}/W =$

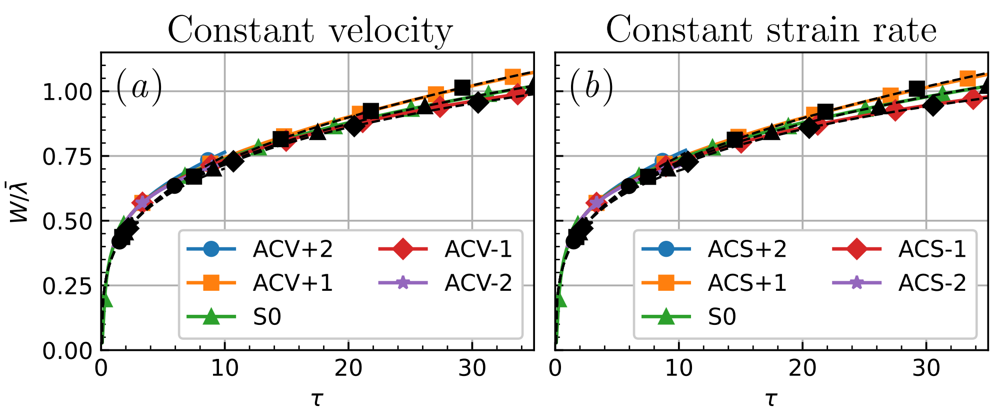


FIG. 16: Buoyancy-drag model for simulations under the applied transverse strain rate with (left) constant velocity profile and (right) constant strain rate. Solid line indicates ILES, dashed line indicates buoyancy-drag model from Eq. 53.

0.383, such that a value of $C_b = 0.622$ achieves $\alpha = 0.05$, whilst a value of $C_b = 1$ will achieve a larger growth rate of $\alpha = 0.08$ for the mixing layer growth. In this formulation, the buoyancy-drag model does not reflect the calibration used for the RANS coefficients.

F. Generalisation of transverse closure

The bulk compression closure for the turbulent length scale can be generalised to other two-equation RANS models, such as for the K-ϵ model which transports the turbulent kinetic energy dissipation rate or the K-ω which transports the specific dissipation rate. This transformation is outlined in Appendix A. A complication for the transverse closure is determining the transverse strain rate for more complicated case configurations. For turbulent mixing layers in standard configurations with clear distinctions between the axial direction (normal to the interface) and transverse direction (parallel to interface or plane of homogeneity), identification of the transverse strain rate is a simple endeavour. For more complicated mixing layers and flow structures, the transverse direction may not be uniform in space or time. For example, the tilted rocket rig [83–86], which experiences anisotropic strain, is a popular validation case for RANS models as it is difficult to simultaneously capture the mixing layer width, turbulent kinetic energy, turbulent mass flux velocity and density-specific volume correlation [58, 87]. To account for this, the transverse direction can be considered to be the homogenous plane and orthogonal to the direction of mixing. The direction of mixing can be determined by using the direction of the gradient vector for some quantity that varies through the mixing layer, such as the species mass fraction or turbulent kinetic energy. With this method, the local transverse strain rate can be calculated according to,

$$S_T = \frac{1}{n-1}\left[\begin{pmatrix}1\\1\\1\end{pmatrix} - \frac{\vec{\nabla}\phi}{|\vec{\nabla}\phi|}\right]\cdot\begin{pmatrix}\partial u_1/\partial x_1\\ \partial u_2/\partial x_2\\ \partial u_3/\partial x_3\end{pmatrix}, \tag{55}$$

where n is the number of dimensions of the true flow. The variable to determine the direction of mixing, ϕ, may vary depending upon the case however the turbulent kinetic energy would likely remain useful in all contexts. As the conducted simulations herewithin are performed as one-dimensional but represent three-dimensional flows ($n = 3$), the gradient is evaluated in the x_1 direction, and the evaluated transverse strain rate from equation (55) reduces to the form used in equation (34). A two-dimensional flow however would not contribute a strain-rate in the third direction, hence the modified scaling factor.

The same closure for the bulk compression of the turbulent length scale should be valid for alternative RANS models that include additional equations, such as equations for the turbulent mass flux velocity, scalar variance, or density self-correlation. Two points of future investigation would be for the treatment of length scales in two length scale models, where the dissipation and transport length scales may be most accurately modelled with different bulk compression closures, and for Reynolds stress models which can more accurately predict the anisotropy of the turbulent kinetic energy.

IV. CONCLUSION

The ability of the K-L turbulence model to predict the growth of a Richtmyer-Meshkov induced turbulent mixing layer under anisotropic strain was investigated through a series of simulations. Based upon the quarter-scale θ-group case [74], the strained ILES of Refs. [19, 20] served as the basis for comparison. Three sets of simulations were conducted, each utilising a different closure for the treatment of the turbulent length scale under bulk compression of the fluid, a term that is not theoretically defined for anisotropic strains. The three closure approaches correspond to scaling the turbulent length scale with the axial strain rate, the transverse strain rate, and the default methodology of scaling with the mean strain. Under mean axial strain rates, the K-L model was able to predict the expansion/compression of the mixing layer due to the ability to resolve the velocity difference in the fluid flow which expands/compresses the mixing layer. The results for each closure differed in their ability to model the turbulent growth and the total turbulent kinetic energy, with the transverse strain closure performing the best of the three options. In a similar fashion, for transverse strain application the transverse strain closure performed the best of the three methods for both the integral width and turbulent kinetic energy. These results suggest that for modelling of turbulent mixing layers under anisotropic strain rates, such as observed in implosion and explosion profiles in spherical geometries, the performance of the K-L model can be improved through the modification of the bulk compression term for the turbulent length scale to instead vary with the transverse strain rate. A self-similarity analysis of the K-L model produced a buoyancy-drag model for modelling the mixing layer growth. The development of the model was limited by the assumption of the proportionality of the turbulent length scale and integral width, a common assumption for model calibration in K-L type models. To rectify this constraint, a three-equation buoyancy-drag model was developed, which evolves both the integral width and a turbulent length scale. With this design, the integral width is allowed to scale with the axial strain which stretches/compresses the mixing layer, whilst the turbulent length scale scales with the transverse strain rate. Further avenues of research should be conducted to determine the proficiency of these modifications in describing other flow features.

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Appendix A: Equivalence of two-equation models

Starting from the turbulent kinetic energy and the turbulent length scale, alternate formulations are obtained by using the transformation [9, 88]

$$Z = C_Z K^m L^n. \tag{A1}$$

Given the formulation of the dissipation of turbulent kinetic energy in Eq. 14e, for $Z = \epsilon$ the values of $m = 3/2$, $n = -1$, and $C_Z = 2\sqrt{2}C_D$ are obtained. The transport equation for Z is then obtained via

$$\bar{\rho}\frac{\tilde{D}Z}{\tilde{D}t} = m\frac{Z}{\tilde{K}}\bar{\rho}\frac{\tilde{D}\tilde{K}}{\tilde{D}t} + n\frac{Z}{L}\bar{\rho}\frac{\tilde{D}L}{\tilde{D}t} \tag{A2}$$

$$\begin{aligned} &= m\frac{Z}{\tilde{K}}\left[\frac{\partial}{\partial x_j}\left(\frac{\mu_T}{N_K}\frac{\partial K}{\partial x_j}\right) + \bar{\tau}_{ij}\frac{\partial \tilde{u}_i}{\partial x_j} + S_K - \bar{\rho}\epsilon\right] \\ &\quad + n\frac{Z}{L}\left[\frac{\partial}{\partial x_j}\left(\frac{\mu_T}{N_L}\frac{\partial L}{\partial x_j}\right) + C_L\bar{\rho}\tilde{V} + \bar{\rho}LS_\phi\right] \end{aligned} \tag{A3}$$

where $\tilde{D}/\tilde{D}t = \partial/\partial t + \tilde{u}_i\partial/\partial x_i$ refers to the total derivative for the Favre-averaged flow field. For the turbulent dissipation rate, the equations evaluate to

$$\begin{aligned} \bar{\rho}\frac{\tilde{D}\epsilon}{\tilde{D}t} = &\frac{\partial}{\partial x_j}\left(\mu_T\left(\frac{3}{2N_K} + \frac{1}{N_L}\right)\frac{\partial \epsilon}{\partial x_j}\right) + \frac{3}{2}\frac{\epsilon S_K}{K} + \frac{3}{2}\frac{\epsilon}{\tilde{K}}\bar{\tau}^{(d)}_{ij}\frac{\partial \tilde{u}_i}{\partial x_j} \\ &- \left(\frac{3}{2} + \frac{C_L}{2C_D}\right)\bar{\rho}\frac{\epsilon^2}{\tilde{K}} - \left(\frac{3}{2}C_P\frac{\partial \tilde{u}_i}{\partial x_i} + S_\phi\right)\bar{\rho}\epsilon + C.D. \end{aligned} \tag{A4}$$

On the right-hand side of Eq. A4, the terms represent diffusion, shear production, buoyancy production, dissipation, bulk compression, and cross-diffusion respectively for the turbulent dissipation rate. The re-arrangement of the diffusion terms for $\tilde{K}$ and L into the diffusion term for ϵ introduces cross-diffusion terms [9], which will depend upon the gradients of $\tilde{K}$, ϵ and density, but not the strain rates of the flow. For simplicity, these terms have not been included as the focus of this comparison is on the bulk compression. The Reynolds stress in the shear production term has been split into the deviatoric component $\bar{\tau}^{(d)}_{ij}$ and the isotropic component $-C_P\bar{\rho}K\delta_{ij}$. The isotropic component of the Reynolds stress combines with the bulk compression term for the turbulent length scale to produce the bulk compression term for the turbulent dissipation rate. For comparison, the transport equation for the turbulent dissipation rate in Refs. [10, 11, 89–92] is

$$\bar{\rho}\frac{\tilde{D}\epsilon}{\tilde{D}t} = \frac{\partial}{\partial x_j}\left[\left(\bar{\mu} + \frac{\mu_t}{\sigma_\varepsilon}\right)\frac{\partial \varepsilon}{\partial x_j}\right] + C_{\varepsilon 0}\frac{\varepsilon}{K}S_K + C_{\varepsilon 1}\frac{\varepsilon}{K}\tau^{(d)}_{ij}\frac{\partial \tilde{u}_i}{\partial x_j} - C_{\varepsilon 2}\frac{\rho\epsilon^2}{K} - \frac{2}{3}C_{\varepsilon 3}\rho\varepsilon\frac{\partial \tilde{u}_j}{\partial x_j} \tag{A5}$$

where the form of S_K varies depending on the model, a physical viscosity $\bar{\mu}$ has been included in the diffusion term, and $\bar{\tau}_{ij}$ has been updated changed to use current notation. The values of σ_ϵ and $C_{\epsilon j}$ are model constants. The last term on the right-hand-side of Eq. A5 is the bulk compression term for the dissipation rate. For incompressible flows it is common to set the corresponding coefficient $C_{\epsilon 3}$ to zero, whilst for compressible flows a value of $C_{\epsilon 3} = 2$ is used to achieve the coefficient of $-4/3$ [64, 93]. Using the default model coefficients for the K-L model with $C_P = 2/3$ and $S_\phi = \nabla \cdot \tilde{\mathbf{u}}/3$, the resulting bulk compression term for the turbulent dissipation rate is

$$-\left(\frac{3}{2}C_P\frac{\partial \tilde{u}_i}{\partial x_i} + S_I\right)\bar{\rho}\epsilon = -\frac{4}{3}\frac{\partial \tilde{u}_i}{\partial x_i}\bar{\rho}\epsilon, \tag{A6}$$

giving the same expression as used for K-ϵ models. The model in Ref. [94] neglects the contribution of the isotropic production from K in the derivation for the bulk compression closure of ϵ, instead attaining $C_{\epsilon 3} = 1/2$. For inhomogeneous turbulence with anisotropic strain, the transverse strain closure model for the bulk compression of the turbulent length scale performed better than the default closure for isotropic strain. Using the transverse strain closure, the expression for the bulk compression of the turbulent dissipation rate becomes

$$-\left(\frac{3}{2}C_P\frac{\partial \tilde{u}_i}{\partial x_i} + S_T\right)\bar{\rho}\epsilon = -\left(\frac{\partial \tilde{u}_i}{\partial x_i} + S_T\right)\bar{\rho}\epsilon. \tag{A7}$$

This expression is very similar, in essence swapping out a mean strain rate contribution (one third of the divergence) for the transverse strain contribution. Under isotropic strain, the expressions are identical in the low-Atwood number limit. The transformation process can be repeated for the K-ω model, using $m = 1/2$, $L = -1$, and leaving C_ω as an arbitrary constant. The equivalent transport equation for the specific dissipation rate ω is

$$\begin{aligned}\bar{\rho}\frac{\tilde{D}\omega}{\tilde{D}t} &= \frac{\partial}{\partial x_j}\left(\mu_T\left(\frac{1}{2N_K} + \frac{1}{N_L}\right)\frac{\partial \omega}{\partial x_j}\right) + \frac{1}{2}\frac{\omega S_K}{K} + \frac{1}{2}\frac{\omega}{\tilde{K}}\bar{\tau}_{ij}^{(d)}\frac{\partial \tilde{u}_i}{\partial x_j} \\ &- \left(\frac{C_D^{3/2}2^{1/2}}{C_\omega} + \frac{C_L 2^{1/2}}{2C_\omega}\right)\bar{\rho}\frac{\omega^2}{\tilde{K}} - \left(\frac{1}{2}C_P\frac{\partial \tilde{u}_i}{\partial x_i} + S_\phi\right)\bar{\rho}\omega + C.D.\end{aligned} \tag{A8}$$

The terms on the right-hand-side of Eqn. A8 are presented in the same order as for Eqn. A4. Evaluation of the bulk compression term with the default isotropic closure gives a scaling coefficient of $-2/3$. The 1988 Wilcox K-ω model [95] is given by

$$\bar{\rho}\frac{\tilde{D}\omega}{\tilde{D}t} = \frac{\partial}{\partial x_j}\left[\left(\mu + \frac{\mu_T}{\sigma_\omega}\right)\frac{\partial \omega}{\partial x_j}\right] + \gamma\frac{\omega}{\tilde{K}}\bar{\tau}_{ij}^{(d)}\frac{\partial \tilde{u}_i}{\partial x_j} - \beta\rho\omega^2 - \gamma\frac{2}{3}\bar{\rho}\omega\frac{\partial \tilde{u}_k}{\partial x_k}. \tag{A9}$$

The terms on the right-hand-side are diffusion, shear production, dissipation, and a bulk compression term which has been extracted from the shear production. This model does not have any buoyancy production, and it does not include the cross-diffusion term for simplicity. The coefficient $\gamma = 13/25$ is calibrated for wall properties, but is very close to the derived factor of 1/2 from the model equivalence. The bulk compression term in this K-ω model will then evaluate to around $-1/3$. Based off Morel & Mansour's derivation, Refs. [96, 97] adjust the coefficient by assuming that one-dimensional rapid compression from shock waves is the primary activation of the bulk compression. This results in a scaling of the divergence by $-4/3$ as compared to the isotropic compression closure of $-2/3$. Using the transverse closure will result in the relation

$$-\left(\frac{1}{2}C_P\frac{\partial \tilde{u}_i}{\partial x_i} + S_T\right)\bar{\rho}\omega = -\left(\frac{1}{3}\frac{\partial \tilde{u}_i}{\partial x_i} + S_T\right)\bar{\rho}\omega \tag{A10}$$

An interesting inclusion in the K-ω model is the 'Pope correction' [9, 98], which modifies β in the dissipation term of ω. This modification was created due to the anomaly of the different spreading rates of the round-jet and plane-jet. The plane-jet is a two-dimensional problem, whilst the round-jet is three-dimensional and will experience vortex stretching. The correction introduces a dependence upon χ, a non-dimensional measure of the vortex stretching. Whilst full details can be found in the aforementioned references, the non-dimensional measure for the vortex stretching in the K-ω model is given by

$$\chi_\omega = \left|\frac{\Omega_{ij}\Omega_{jk}\hat{S}_{ki}}{(\beta^*\omega)^3}\right| \tag{A11}$$

For the strained simulations conducted, this correction would not contribute as χ would be zero due to the dependence on the rotation tensor of the flow.

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